

Violence Reduction, Firms, and Local Labor Markets: Evidence from a Ceasefire

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Abstract

While violence and under-development often coexist, the economic consequences of reduced violence for local economic activity remain poorly understood. I study El Salvador's 2012 gang truce, which coincided with a sharp 60% drop in homicides, to examine how improvements in security relate to employment and firm activity. Using administrative data on firms and homicides, I compare areas with higher measured pre-truce gang exposure to areas with lower or unobserved measured exposure. Employment and the number of firms increase after the truce, with the strongest responses concentrated among small and medium-sized firms: employment rises by roughly 6 percent and the number of firms by about 3 percent. The pattern of results is most consistent with stronger local demand and less constrained local operation, reflected in fewer firm closures, more openings, stronger responses in non-tradable sectors, and supportive evidence from wages and consumer confidence. These gains weaken as violence resurges, highlighting the fragility of temporary security improvements.

Keywords: Organized crime, homicides, labor markets, local demand, firms, economic growth

JEL codes: D74, K42, O17

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1 Introduction

A central question in economics is how violent crime shapes economic activity. Existing evidence has made substantial progress on the costs of conflict, terrorism, and organized crime, but less is known about how local economies respond when lethal violence falls sharply, especially outside the set of listed, exporting, or manufacturing firms most often observed in firm-level studies. This gap in the literature is striking given that criminal organizations and violent crime remain widespread. In Latin America and the Caribbean (LAC), for example, homicide rates exceed the global average by more than threefold, and organized criminal groups account for roughly one-half of all killings (United Nations Office on Drugs and Crime (UNODC), 2023). Moreover, 54% of households report the presence of criminal organizations in their neighborhoods (Uribe et al., 2022), raising important questions about their consequences for local economic performance. While much research explores how economic improvements can curb criminal activity,¹ the reverse channel, how violent crime itself affects economic performance, remains poorly understood.

This paper studies the economic effects of a sharp decline in homicides during the 2012 Salvadoran gang truce. Secretly negotiated between incarcerated gang leaders and government officials, the agreement cut homicide rates by an estimated 60% within weeks. The episode provides a rare setting in which a large, sudden improvement in lethal violence can be linked to the response of firms and local labor markets.

Using the truce’s sharp timing and differences in measured pre-truce gang exposure across locations, I document sizable gains in local economic activity in more exposed areas. Employment and the number of firms increase after the truce, with the strongest and most stable responses concentrated among small and medium-sized firms. The baseline estimates imply roughly a 6% increase in employment and a 3% increase in the number of firms; matched and synthetic difference-in-differences estimates remain positive, with firm-count effects especially stable. A complementary specification based on realized homicide declines yields a similar economic message: larger reductions in violence are associated with larger improvements in local employment and firm activity. The pattern of results is most consistent with improved local operating conditions and stronger local de-

¹For early analyses, see Jones (1932) and Simpson (1932), as well as more comprehensive discussions by Bushway and Reuter (2011) and Mustard (2010). Specific economic shocks studied include oil shocks (Raphael and Winter-Ebmer, 2001), changes in industrial composition through Bartik instruments (Gould, Weinberg and Mustard, 2002; Fougère, Kramarz and Pouget, 2009), exchange rate movements (Lin, 2008), international mineral prices (Axbard, Benshaul-Tolonen and Poulsen, 2019), trade liberalization (Dix-Carneiro, Soares and Ulyssea, 2016), mass layoffs and plant closures (Bennett and Ouazad, 2020; Pinotti, Britto and Sampaio, 2020), and improvements in legal market conditions (Pinotti, 2017).

mand: gains are concentrated in non-tradable sectors and accompanied by fewer business closures, more openings, and suggestive evidence from wages and consumer confidence. Together, these patterns are consistent with households and firms reengaging with local markets as security improved, creating conditions for firms to survive, expand, and hire.

The temporary nature of the truce highlights the fragility of these gains. As the agreement deteriorated and violence resurged, the number of firms reverted toward its pre-truce level, meaning that the earlier increase in the stock of firms was largely wiped out. Employment stopped growing relative to its earlier post-truce trajectory, suggesting that while labor markets retained some gains, the underlying growth potential was not sustained. This divergence underscores that preserving economic improvements requires lasting rather than temporary security improvements. Consistent with these local effects, national statistics also show a temporary rise in aggregate labor force participation during the truce years, a pattern not observed in other countries in the region, that faded once violence resurged.

To study how the truce relates to local economic activity, I use a difference-in-differences framework that compares areas with higher measured gang exposure before the truce to areas with lower or unobserved exposure. The main exposure measure is based on whether a polygon recorded at least one gang-related homicide before the truce, capturing places where gang presence was more visible before the agreement. The design therefore identifies how economic activity changed in areas where the truce was more likely to relax local security constraints, relative to less exposed areas. I interpret the estimates as differential responses across areas with different levels of measured pre-truce gang exposure. The main findings are robust to matching estimators that compare more similar polygons and to synthetic difference-in-differences estimates that construct an alternative counterfactual path. I also report a complementary continuous specification based on realized homicide reductions, which helps summarize the gradient between the magnitude of violence declines and local economic improvements.

This paper contributes to the growing literature on how organized crime and violence shape economic outcomes, including stock-market valuations and financial conditions (Abadie and Gardeazabal, 2003, 2019; Guidolin and La Ferrara, 2007, 2010), firm performance and survival (Ksoll, Macchiavello and Morjaria, 2010; Camacho and Rodriguez, 2013; Collier and Hoeffler, 2004; Klappper, Richmond and Tran, 2013; Roza, 2018), and employment and financial stability under criminal capture or infiltration (Fenizia and Saggio, 2024; Mirenda, Mocetti and Rizzica, 2022). It also complements work that quantifies the benefits of crime reduction through perceived valuations (Ludwig

and Cook, 2001; Cook, Jeuland and Ludwig, 2025).

Relative to evidence on institutional removal of organized-crime influence or on the economic costs of rising violence, this paper studies a sudden decline in lethal gang violence and shows that the gains are concentrated in smaller, locally embedded firms and non-tradable sectors.

Closer to the question examined here, some studies have specifically analyzed how violence affects labor market participation and local economic conditions. For instance, Utar (2024) and Velásquez (2020) document that increased violence discourages labor market participation due to heightened fears of victimization, notably among self-employed women who shift labor hours toward household activities, and blue-collar workers in labor-intensive manufacturing firms. Brodeur (2018) likewise shows that successful terror attacks reduce local employment and worsen consumer sentiment relative to failed attacks, pointing to the importance of expectations and perceived safety in shaping economic outcomes.

This paper builds on that literature by studying an unusually large and sudden decline in lethal violence. The Salvadoran truce generated a sharp and sustained reduction in homicides during the truce period, providing an informative setting to study how sizable improvements in security relate to employment and firm activity.

In addition to exploiting this distinctive shock, the analysis covers a broad cross-section of formal local economic activity rather than subsets of firms with high productivity or international exposure. Prior studies typically focus on stock market-listed firms, exporters, or specific manufacturing industries, while this paper examines broader effects across sectors and firm sizes. A novel finding is that smaller firms and those in non-tradable sectors, which depend more heavily on local demand, experience the largest gains. This result complements recent evidence by Utar (2024), who shows that within manufacturing, plants that rely more heavily on domestic inputs and markets are especially vulnerable to violence. While that study is limited to one sector, my findings suggest that the economic benefits of improved security extend well beyond manufacturing and are consistent with local demand playing a central role across multiple sectors. This has broader implications for understanding how violence shapes economic activity in high-crime contexts.

Earlier work often shows that increases in crime reduce labor supply, leading to lower employment and upward pressure on wages, consistent with a leftward shift in labor supply (Utar, 2024; Velásquez, 2020). In the Salvadoran case, the sharp decline in violence is followed by higher employment and firm activity, while the wage evidence does not point to wage declines and is suggestive of stronger local labor demand. To interpret these patterns, I develop a simple theoretical framework

in which crime affects both the willingness to work and the willingness of consumers and firms to transact in local markets. The framework clarifies why employment can rise through more than one margin: safer conditions may increase labor supply, but they may also reduce fear and transaction costs in local markets, raising demand for local goods and services. The empirical patterns—fewer closures, stronger non-tradable activity, and suggestive wage and consumer-confidence evidence—are most consistent with local demand and operating conditions being important margins, even if several channels likely moved at the same time.

An additional question is whether the economic gains extended beyond areas with higher measured exposure. Spillovers matter because they may affect the interpretation of the main estimates, but they are also outcomes of independent interest. If neighboring or lower-exposure areas also benefited from improved security, the baseline contrast should be interpreted as a differential-exposure estimate rather than as a pure direct effect. To assess this possibility, I examine whether adjacent polygons also experienced economic improvements. While much of the prior literature highlights negative spillovers—crime being displaced into nearby neighborhoods (Draca, Machin and Witt, 2011; Dell, 2015)—the evidence here is instead suggestive of positive economic spillovers. These exercises are necessarily cautious, but they indicate that the main patterns remain positive when nearby exposure is taken into account.

Although the Salvadoran truce was institutionally distinctive, the margins highlighted by the evidence, including fear costs, local demand, and the dependence of smaller firms on nearby consumers, are relevant in many high-crime settings. This suggests that the mechanisms emphasized here may travel beyond this case, while the estimates should be read in light of the truce's temporary and institutionally specific nature. This interpretation is consistent with broader evidence linking crime to firm performance. For instance, Islam (2014) show that economic activity is negatively correlated with crime across firms in 27 developing countries, with the relationship especially pronounced for small and medium firms. By linking a sharp decline in violence to changes in local employment and firm activity, this paper contributes to a broader understanding of how reducing crime can shape economic outcomes, and under what circumstances such gains are most likely to materialize.

The estimates in this paper likely understate the economic benefits of more durable and broader security improvements. Because the truce was temporary and its credibility uncertain, firms and workers may have postponed investments and hiring that would have occurred under a permanent decline in violence. In addition, the agreement primarily reduced homicides, while other offenses

such as extortion showed no clear decline, and evidence from a later non-aggression pact among the same gangs suggests that extortion may even rise in some contexts.² A more stable and broad-reaching reduction in crime would therefore be expected to generate larger increases in economic activity than those estimated here.

The remainder of the paper is structured as follows. Section 2 provides background on violence in El Salvador and the 2012 gang truce. Section 3 describes the data, and Section 4 presents the empirical framework. Section 5 presents the main findings on employment and firm activity, along with supporting credibility checks. Section 6 examines the margins through which these effects materialize, focusing on firm survival, sectoral composition, wages, and aggregate consumer-confidence indicators. Section 7 uses a simple theoretical framework to interpret the empirical patterns. Section 8 examines spillovers, and Section 9 analyzes the economic reversal as the truce deteriorated and violence rose again. Section 10 concludes.

2 Context

2.1 Violence in El Salvador

El Salvador entered the 2010s with very high levels of violence, much of it linked to gang activity. The country's experience with homicides and extortion was closely tied to the presence of gangs such as MS-13 and Barrio 18. These gangs, formed by Latin American migrants in Los Angeles, expanded after their return to El Salvador in a context of weak institutions and limited law enforcement (Arana, 2005). By 2010, crime had become the country's most pressing issue, with 61% of Salvadorans identifying it as their top concern, surpassing economic issues (35%) (Lab, 2010).

The arrival of deported gang members from the US in the mid-1990s led to the replication of behaviors and structures familiar in the contentious environments of Los Angeles. These individuals formed local chapters of MS-13 and Barrio 18, bringing with them gang symbols, language, and tattoos, and creating a collective identity through violence. Further, this period was characterized by a lack of government programs to address gang violence or reintegrate deported Salvadorans, which enabled a further escalation of the gang crisis.

Data from the National Police indicate that at least one-third of homicides are attributable to gang-related activities. More broadly, crime and violence impose large economic costs, with esti-

²Brown et al. (2024) studies a 2016 non-aggression pact and documents higher reported extortion, drawing on payment records from a logistics company that distributes goods across El Salvador.

mates suggesting losses between 6.5% and 16% of the country's Gross Domestic Product (GDP) (Jaitman et al., 2017; Peñate et al., 2016). Crime has also been a persistent barrier to investment, with 47% of businesses identifying it as the most critical factor affecting their investment decisions in El Salvador in 2010 (FUSADES, 2010–2020). Studies by Melnikov, Schmidt-Padilla and Sviatschi (2020); Kalsi (2018); Castro et al. (2025); Castro and Kotti (2022); Castro et al. (2019) reveal the profound impact of gang presence on a wide spectrum of socio-economic factors within these neighborhoods. Notably, households in these areas face significant challenges concerning income, educational opportunities, housing quality, and electoral participation, all of which are negatively influenced by the pervasive control exerted by gangs.

2.2 Truce Negotiations

In 2012, a major shift occurred when the Salvadoran government facilitated a truce between MS-13 and Barrio 18. Contemporary reporting and subsequent research describe the truce as an initially secret and politically ambiguous process centered on contacts with incarcerated gang leaders (El Faro, 2012; Washington Office on Latin America, 2013; Dudley, 2013; van der Borgh and Savenije, 2019). The concessions most consistently documented at the onset involved prison transfers, less restrictive prison conditions, and expanded visitation privileges for gang leaders, while the government initially denied having negotiated with gangs (El Faro, 2012; Reuters, 2012; Washington Office on Latin America, 2013; Dudley, 2013; van der Borgh and Savenije, 2019). This truce was associated with a roughly 60% reduction in homicides, bringing homicide rates to their lowest levels since the end of the civil war (Washington Office on Latin America, 2013; Dudley, 2013; van der Borgh and Savenije, 2019).

While the truce was successful in reducing homicides, the number of reported extortions remained nearly unchanged compared to pre-truce levels (see Online Appendix Figure B.4) (Dudley, 2013; van der Borgh and Savenije, 2019). The truce is commonly understood to have lasted for approximately 24 months, although its end did not occur on a single sharply defined date. The agreement began to deteriorate as political and institutional support weakened: the Minister of Security associated with the process was removed, and gang leaders were later deprived of the privileges granted during the truce. After this deterioration, the government shifted toward a more confrontational anti-gang strategy, intensifying clashes between security forces and gang members. Homicide rates then surged beyond pre-truce levels, culminating in a rate of 103 per 100,000 inhabitants in 2015, the highest recorded since the civil war.

3 Data

This study relies on two primary administrative data sources, which are aggregated at the local geographic level. The unit of analysis is a set of 1,581 polygonal geographic units covering the entire country. These polygons correspond to small administrative geographic units defined for territorial organization purposes. They provide a more granular partition of the country than municipalities and allow for a finer characterization of local economic conditions, labor markets, firm activity, and violence. Firm and crime data are aggregated to these polygons and observed over time, yielding a monthly panel from 2010 to 2015.

1. **Social Security Database (ISSS):** This administrative dataset provides monthly records of formal firms registered with the Salvadoran Social Security Institute. It includes information on firms' primary activity, number of employees, and total payroll.
2. **National Civil Police Records:** This dataset provides monthly records of reported homicides across the country. I use these records to measure local violence and construct the main proxy for measured pre-truce gang exposure, based on pre-truce homicides classified as related to MS-13 or Barrio 18.

The social security and police records are combined to construct the main polygon-level panel. I also use aggregate indicators from the Economic Outlook Reports published by FUSADES, a Salvadoran economic and social policy think tank, as complementary contextual evidence. These indicators include consumer confidence, business confidence, and firms' perceived obstacles to investment during the truce period.

3.1 Measuring Pre-Truce Gang Exposure

A central challenge in this setting is that gang presence is not directly observed across locations. While police intelligence provides detailed information on criminal activity, there is no comprehensive measure of the geographic footprint of gangs prior to the truce. To address this, I construct a proxy for pre-truce gang exposure using information on homicides related to MS-13 and Barrio 18, the country's two main gangs.

Specifically, I define an indicator variable that takes the value one if a polygon experienced at least one homicide classified in the police records as related to MS-13 or Barrio 18 in the pre-truce

period, and zero otherwise. This measure captures areas with observed gang activity prior to the truce and serves as a proxy for exposure to gang dynamics.

Importantly, this proxy is imperfect. Not all areas with gang presence necessarily record gang-related homicides, either because such events did not occur during the pre-truce window or because they were not classified as gang-related. As a result, some areas with gang exposure may be included in the comparison group. Throughout the analysis, I therefore interpret this variable as capturing differences in *measured* exposure to gangs rather than a sharp distinction between treated and untreated locations.

This interpretation has two implications. First, the empirical design compares areas with higher measured exposure to areas with lower or unobserved exposure, rather than fully exposed versus unexposed areas. Second, to the extent that gang-exposed areas are misclassified into the comparison group, the estimated differences may be attenuated if misclassified gang-exposed areas responded more like higher-exposure polygons than like truly lower-exposure areas.

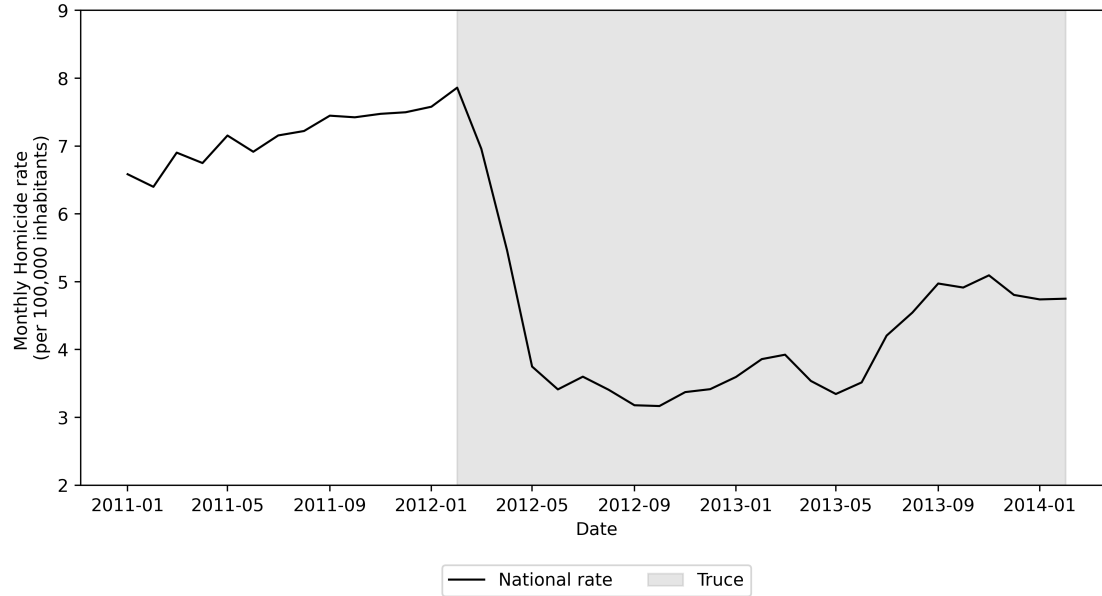
Areas with higher measured exposure differ from lower-exposure areas in baseline economic scale, as shown in Appendix Table A.1. The baseline specification retains the full set of polygons, including areas with little or no pre-truce formal economic activity. Because the empirical design relies on within-polygon changes over time, time-invariant differences in levels are absorbed by unit fixed effects. The natural remaining concern is whether baseline economic scale is associated with differential outcome trends or differential responsiveness to the truce.

I address this concern using three complementary checks. First, I use event-study specifications to examine whether higher- and lower-exposure areas followed similar pre-truce trends. Second, I report matched-sample estimates that restrict comparisons to polygons with more similar pre-truce economic scale. Third, I report synthetic difference-in-differences estimates that construct an alternative counterfactual path. Together, these checks assess whether the main patterns persist after focusing on more comparable outcome dynamics and comparison groups.

3.2 Violence Trends and the Truce

The truce coincided with a sharp national decline in homicides. Figure 1 illustrates this pattern, showing a pronounced drop in homicide rates at the national level following the onset of the truce.

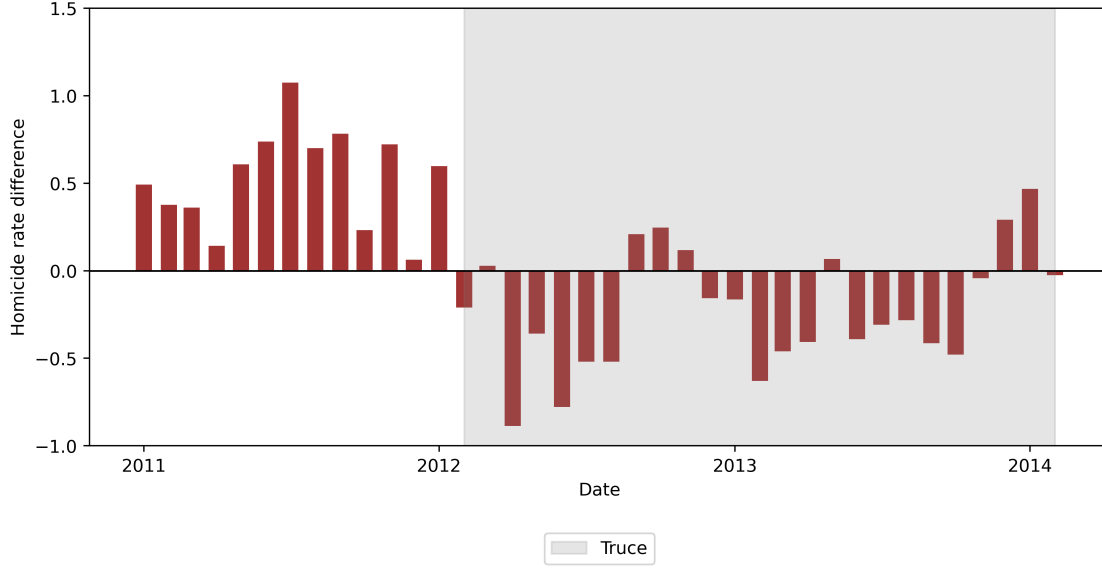
Figure 1: National Homicide Trends Around the Truce



Notes: This figure plots the national monthly homicide rate before and after the onset of the truce. The series is smoothed using a trailing 3-month moving average. The vertical line indicates the start of the truce period.

Consistent with the national scope of the truce, homicide rates decline in both higher- and lower-exposure areas, as shown in Appendix Table A.1 and in the national series in Figure 1. Areas with higher measured gang exposure exhibit relatively larger reductions. Figure 2 summarizes this relative contrast by showing how the homicide-rate gap between higher- and lower-exposure areas evolves over time.

Figure 2: Difference in Homicide Rates by Pre-Truce Gang Exposure



Notes: This figure plots the monthly difference in homicide rates between polygons with at least one gang-related homicide prior to the truce and those without. The bars summarize how the gap in violence narrows after the onset of the truce, consistent with larger declines in areas with higher measured gang exposure.

These patterns are consistent with the interpretation that the truce changed homicide dynamics at the national level while generating larger local security improvements in areas with greater measured pre-truce gang exposure. In this sense, the proxy for gang exposure captures variation in differential exposure to the truce-related decline in lethal violence.

4 Empirical Framework

The primary empirical framework employs a standard two-way fixed effects (TWFE) difference-in-differences (DiD) framework. I use the common timing of the truce together with cross-polygon differences in measured pre-truce gang exposure. Under this approach, $D_i \in \{0, 1\}$ is a binary indicator equal to one if polygon i recorded at least one gang-related homicide before the truce, and zero otherwise. The estimand is a differential response by measured exposure to the truce-related improvement in local security conditions, rather than the total effect of the truce relative to unaffected areas. As discussed above, this variable should be interpreted as a proxy for pre-truce gang exposure rather than as a clean treatment assignment. It captures areas with higher observed gang presence prior to the truce, while areas with $D_i = 0$ should be interpreted as polygons with

lower or unobserved measured exposure.

This historical context shapes the interpretation of the design. The truce was not announced *ex ante* as a public security or economic policy; it became public through press reporting after the homicide decline had begun. The historical record does not point to a contemporaneous public economic, labor-market, or business-formalization policy targeted to higher-exposure areas at the onset of the truce. The contemporaneous concessions most consistently documented concern incarcerated gang leaders, especially prison transfers, less restrictive prison conditions, and expanded visitation privileges (El Faro, 2012; Reuters, 2012; Washington Office on Latin America, 2013; Dudley, 2013; van der Borgh and Savenije, 2019). I therefore interpret the estimates as differential local responses to truce-related security conditions, whose most visible and empirically central component was the sharp decline in lethal violence, rather than as the effect of a cleanly isolated policy instrument.

Because the exposure proxy is measured with error, the estimates may be attenuated relative to the true contrast between areas with higher and lower gang exposure. I therefore interpret the coefficients as comparisons across levels of measured exposure. The central comparison assumption is that, absent the truce-induced change in violence, outcomes in high- and low-exposure areas would have followed parallel trends. I assess the plausibility of this assumption using a standard event-study specification:

$$y_{i,t} = \alpha_i + \gamma_q + \sum_{q \neq 0} \delta_q (D_i \times \mathbf{1}\{Q_t = q\}) + \varepsilon_{i,t}, \quad (1)$$

where $y_{i,t}$ is the outcome for unit i at time t , α_i are unit fixed effects, and γ_q are event-time period fixed effects. D_i is the time-invariant indicator for measured pre-truce gang exposure, equal to one if polygon i recorded at least one gang-related homicide before the truce. Q_t indexes three-month event-time intervals relative to the truce, with $Q_t = 0$ denoting the omitted interval immediately preceding the truce. Each coefficient δ_q measures the differential outcome path of higher-exposure polygons relative to lower- or unobserved-exposure polygons, normalized to that omitted pre-truce interval. By examining the coefficients $\hat{\delta}_q$ for periods prior to the truce, I find no evidence of differential pre-trends across areas with higher and lower measured exposure, with estimates that are sufficiently precise to rule out economically meaningful differences. In the results section, I also report matching and synthetic difference-in-differences (SDID) estimates as robustness checks for this main specification.

Although the binary exposure proxy constitutes the main empirical design, I also report a com-

plementary specification that uses variation in the magnitude of the realized homicide decline during the truce.

Specifically, I construct a continuous measure of treatment intensity based on the reduction in homicides observed in each polygon during the truce period. This measure captures the idea that, although the homicide decline was visible nationally, its magnitude varied substantially across locations. I implement the continuous exercise using a residualized dose-response specification described in Online Appendix Section B.1.1.

This specification is best interpreted as descriptive evidence on how economic outcomes covary with the magnitude of violence reductions, rather than as a standalone causal identification strategy. It provides a characterization of how economic outcomes vary with the magnitude of violence reductions across locations, complementing the main binary exposure design.

In the results section, I report estimates based on linear and quadratic specifications of treatment intensity, as well as event-study evidence that describes outcome dynamics across areas with larger and smaller realized homicide reductions.

4.1 Preview of Supporting Checks

I briefly describe the supporting checks that are later reported in the results section; more detailed derivations are provided in the Online Appendix B.1.

First, for the main measured-exposure specification based on pre-truce gang-related homicides, I report matched-sample estimates that compare areas with higher measured exposure to observably similar areas with lower measured exposure. This helps assess whether the results are sensitive to the composition of the comparison group.

Second, I report synthetic difference-in-differences (SDID) estimates for the same main specification. This provides an additional way to construct the counterfactual path and assess whether the main patterns are robust to a different weighting scheme.

Third, for the complementary continuous specification based on realized homicide declines, I report two supporting checks. First, I examine whether realized homicide reductions are systematically predicted by pre-truce economic conditions. Second, I report placebo exercises that assign larger- versus smaller-decline status in periods before the truce and consider alternative intensity thresholds. These checks help characterize whether the descriptive gradient is related to pre-truce economic growth measures or appears similarly when the homicide-reduction measure is assigned to placebo periods.

5 Main Results

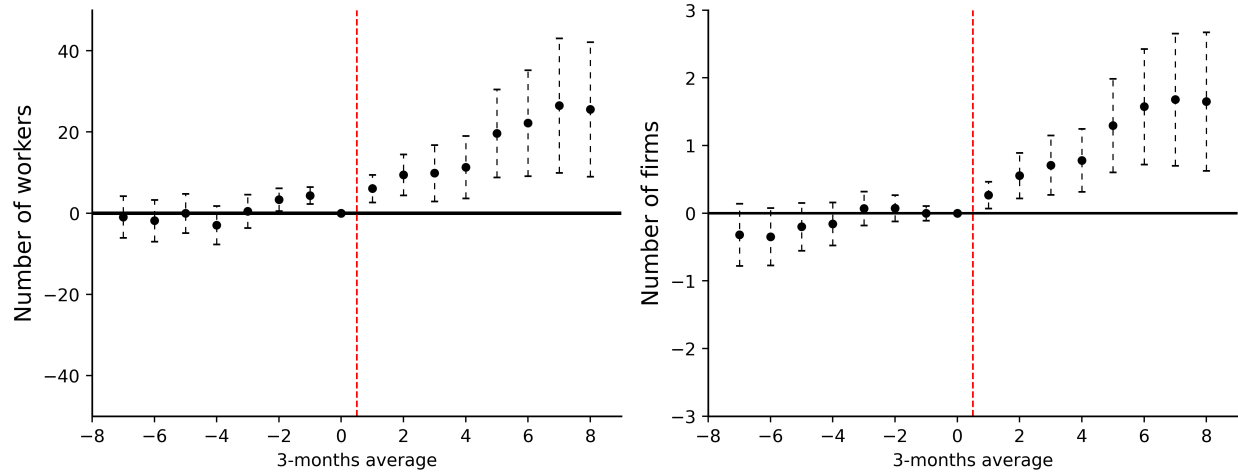
This section presents the main empirical findings on the economic effects of the truce. I begin by documenting the differential evolution of economic outcomes across areas with different levels of measured pre-truce gang exposure. I then assess the robustness of these patterns to alternative comparison strategies, and finally examine how outcomes vary with the magnitude of violence reductions across locations.

5.1 Main Effects of the Truce

I begin by examining how economic outcomes evolved following the truce across areas with different levels of pre-truce gang exposure. The analysis compares polygons with at least one observed gang-related homicide before the truce to polygons with lower or unobserved measured gang exposure.

Figure 3 presents event-study estimates for the number of workers and the number of firms across these groups, focusing on small and medium firms (bottom 95%). The corresponding results for the full sample are reported in Appendix Figure A.1. Before the truce, the estimates do not show a sharp break around the onset of the truce. After the onset of the truce, the two outcomes begin to diverge. Employment increases relatively quickly in higher-exposure areas and remains elevated throughout the post-truce window, while the number of firms also rises in a similar pattern. Taken together, the event-study pattern is consistent with a relative improvement in local economic activity in areas with greater pre-truce gang exposure.

Figure 3: Event-Study Estimates for Workers and Firms, Excluding the Top 5% Largest Firms



Notes: This figure displays event-study estimates from Equation (1). The X-axis consolidates data into three-month intervals, covering 24 months before and after the onset of the truce. The omitted category is the three-month interval immediately preceding the truce. All regressions include unit and time fixed effects, and the sample excludes the top 5% of firms by size. Vertical bars indicate 95% confidence intervals, with standard errors clustered at the geographic unit level.

The magnitude of these effects is economically meaningful. In the post-truce period, polygons with higher pre-truce exposure exhibit sustained increases in both employment and firm counts relative to lower-exposure areas. Table 1 reports the corresponding difference-in-differences estimates. For small and medium firms (bottom 95%), the estimates imply an increase of about 16 workers per polygon (roughly 6 percent relative to the pre-truce mean) and about 1.2 additional firms (roughly 3 percent), both statistically significant. By contrast, the coefficients for the top 5% of firms are small and statistically indistinguishable from zero, indicating little response among the largest establishments. When all firms are pooled together, the estimates remain positive: employment increases by about 20 workers per polygon (roughly 2.7 percent, marginally significant), and the number of firms rises by about 1.17 (roughly 2.9 percent, statistically significant). Overall, the magnitudes are stable in sign across samples, but the economically meaningful and statistically stronger responses are concentrated among the bottom 95%.

Table 1: Main Difference-in-Differences Estimates for Employment and Firms

Variables	Workers			Number of firms		
	Bottom 95%	Top 5%	All	Bottom 95%	Top 5%	All
Treatment	16.027*** (5.747)	3.988 (6.478)	20.014* (10.426)	1.175*** (0.366)	-0.008 (0.016)	1.167*** (0.369)
Mean	267.543	463.082	730.624	38.314	2.300	40.614
R-squared	0.9958	0.9967	0.9973	0.9986	0.9993	0.9988
Obs.	25296	25296	25296	25296	25296	25296
Clusters	1581	1581	1581	1581	1581	1581

*Notes: This table reports difference-in-differences estimates of differential post-truce changes in employment and the number of firms across areas with higher versus lower measured pre-truce gang exposure. Estimates are obtained from a two-way fixed effects (TWFE) specification analogous to Equation (1). Results are shown separately for small and medium firms (bottom 95%) and large firms (top 5%). All regressions include unit and time fixed effects. Standard errors, clustered at the geographic unit, are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Taken together, these findings indicate that areas with greater measured pre-truce gang exposure experienced meaningful relative improvements in local economic activity after the truce. The effects are concentrated among small and medium firms, where both employment and firm counts increase, while large firms show little detectable adjustment. This pattern is consistent with improvements in local security conditions primarily benefiting smaller, more locally embedded establishments.

5.2 Addressing Comparability

A central concern in this setting is that areas with gang presence prior to the truce may differ systematically from other areas in terms of their underlying level of economic activity. In particular, polygons with higher measured gang exposure tend to be larger and more economically active, as reflected in higher numbers of firms and workers before the truce. If more economically active areas also exhibit stronger growth dynamics, the baseline estimates could partly capture differences in scale rather than differential responses to the reduction in violence.

To address this concern, I implement a matched-sample design that restricts comparisons to areas with similar levels of pre-truce economic activity. Specifically, I match higher-exposure polygons to lower- or unobserved-exposure polygons based on the number of firms observed prior to the truce. This variable provides a direct proxy for the local scale of economic activity, capturing the size of the local business base. As shown below, matching on this dimension also improves balance

across several other baseline characteristics, although I do not interpret the matched sample as fully balanced on all determinants of growth.

I denote these specifications as $NN(k)$, where k is the number of nearest neighbors used to construct the matched comparison group. The preferred matched specification, $NN(5)$, uses five-nearest-neighbor caliper matching on pre-truce firm counts and matches without replacement. This is a restrictive design: each comparison polygon can enter at most once, so the estimates do not rely heavily on repeated use of a small number of comparison areas. The purpose of this exercise is to ask whether the results persist within a region of more comparable pre-truce economic scale.

Table A.2 documents these balance improvements. Differences in pre-truce firm counts fall substantially, and the matched sample is also closer in employment, wages, opening and closure rates, and baseline homicide rates. Residual differences remain, so I interpret the matched estimates as evidence from a more comparable subset rather than as a design that balances all possible determinants of growth. As is typical with common-support restrictions, the matched estimates pertain to treated polygons for which close comparison areas are available.

Figure A.2 and Table A.3 present the corresponding matched difference-in-differences estimates. The estimates remain positive and statistically significant, consistent with larger post-truce increases in areas with higher measured gang exposure, even among polygons with more similar pre-truce economic activity. The matched event-study estimates also show no evidence of differential pre-truce dynamics within this more comparable subset.

While the matched estimates are smaller in levels than the baseline TWFE estimates, their magnitude is more informative when expressed relative to the matched-sample means. Under the preferred $NN(5)$ design, the employment coefficient of 2.33 workers corresponds to an increase of about 3.3 percent relative to the matched-sample mean, while the firm-count coefficient of 0.47 corresponds to roughly 4 percent. Thus, the smaller level effects appear to reflect the lower economic scale of the matched sample rather than the disappearance of the response.

To assess robustness, I also consider $NN(10)$ and $NN(15)$ specifications that use larger neighborhoods, tighter calipers, and allow limited replacement of comparison polygons. As shown in Online Appendix Tables B.2 and B.3, these alternatives achieve closer balance not only in pre-truce firm counts, but also across the other baseline level characteristics; Online Appendix Table B.4 shows that the implied effects remain similar when expressed as coefficient-to-mean ratios: across the $NN(5)$, $NN(10)$, and $NN(15)$ specifications, employment increases by roughly 2.5–3.3 percent, while firm counts increase by about 4 percent. The firm-count estimates are particularly stable and

statistically significant across all specifications; employment effects are somewhat more sensitive in precision, but remain positive.

Overall, the matched-sample analysis shows that the main patterns remain visible within areas that are more comparable in pre-truce economic scale. The estimates are smaller in levels, as expected given the lower economic scale of the matched sample. When expressed relative to matched-sample means, the firm-count effects remain very close to the baseline magnitude, while the employment effects are smaller but still positive and economically meaningful.

5.3 Alternative Counterfactual

While the event-study evidence suggests that differential pre-trends are limited, a remaining concern is that areas with higher and lower measured exposure may differ in more subtle ways in their underlying trends. If so, standard difference-in-differences estimates could partly reflect differences in pre-existing dynamics rather than post-truce divergence across exposure groups.

I therefore report a synthetic difference-in-differences (SDID) exercise, which constructs an alternative counterfactual by weighting lower- or unobserved-exposure areas to better match the pre-truce evolution of higher-exposure areas. This provides an alternative counterfactual path that places more weight on comparison areas with similar pre-truce outcome dynamics.

Figure A.4 presents the corresponding event-study estimates. The key result is that post-truce divergence remains visible under this alternative weighted counterfactual, which is constructed to better match pre-truce outcome dynamics. Table A.3 reports the associated average effects, which remain positive and economically meaningful, though somewhat smaller for employment.

Taken together, these findings suggest that the main patterns are not solely explained by differences in underlying trends. Even under a more demanding counterfactual that closely matches pre-truce dynamics, areas with higher measured exposure show meaningful post-truce improvements in employment and firm activity.

5.4 Continuous Variation in Violence Reductions

To better describe the magnitude of the economic response, I use continuous variation in the decline in violence across areas. Specifically, I construct a standardized measure of the reduction in homicides following the truce, allowing me to characterize how local economic outcomes vary with a one standard deviation larger decline in violence. The technical details of this complementary

specification are described in Online Appendix Section B.1.1. This approach is primarily descriptive and complements the baseline identification strategy, providing insight into the magnitude and variation of the economic response.

Table A.4 reports the corresponding regression estimates. A one standard deviation decrease in violence is associated with an increase of about 8.4–8.5 percent in employment and about 4.6 percent in the number of firms after one year. The implied associations grow over the two-year horizon, reaching roughly 17 percent for employment and close to 9 percent for firms. The corresponding event-study estimates, shown in Appendix Figure A.5, do not indicate differential pre-truce movements under the linear continuous specification. These magnitudes are economically meaningful and consistent with the patterns from the baseline difference-in-differences results.

A key concern with this complementary exercise is that areas experiencing larger realized declines in violence may also differ in underlying growth dynamics. To assess this, I group areas according to the magnitude of violence reductions and estimate event-study specifications for each group. Figure A.6 shows broadly similar pre-truce movements across intensity groups and a clearer post-truce gradient, with larger improvements in areas experiencing larger realized declines in violence.

I further examine whether the magnitude of the decline in violence is systematically related to pre-existing economic conditions. Table A.5 shows that population size and pre-truce violence are the most consistent predictors of exposure to homicide reductions. By contrast, baseline firm counts, employment, wages, and pre-truce growth are not consistently robust predictors once these demographic and violence controls are included. This helps address the concern that realized homicide reductions are mainly selecting polygons with stronger pre-truce economic growth dynamics.

Finally, I use related placebo exercises based on the homicide-reduction classification described in Online Appendix Section B.2.1. These checks assign areas to larger- versus smaller-decline groups in placebo periods before the truce, and also consider alternative intensity thresholds that more closely mirror the dose-response exercise. Online Appendix Tables B.8 and B.9 show no systematic relationship between placebo assignments and economic outcomes outside the truce period, while the relationship becomes positive during the actual truce period.

Taken together, these results show a positive gradient between realized homicide reductions and local economic activity. The supporting checks indicate that similar patterns are not present in placebo periods and that the realized reductions are not strongly predicted by pre-truce economic growth measures.

6 Mechanisms: Firm Survival, Local Demand, and Labor Markets

The previous section shows that areas with higher measured pre-truce gang exposure experienced larger post-truce gains in employment and firm counts, especially among small and medium-sized firms. These results raise a question about the type of adjustment behind the gains. Higher employment alone is not diagnostic: safer conditions can increase labor supply by making work less costly, while also increasing local demand by making households and firms more willing to transact locally.

This distinction guides the mechanism analysis. If the response is dominated by improved local operating conditions and local demand, one would expect fewer firm closures, stronger activity in non-tradable sectors, and improved consumer and firm perceptions. Wage responses are also conceptually informative: a response driven only by greater labor availability would put downward pressure on wages, while stronger local demand could offset or dominate that force. The evidence below does not identify a single mechanism. Instead, it asks which margins appear most salient in the data. The clearest patterns are firm survival and stronger local-facing activity; wages and aggregate confidence indicators provide supporting but less definitive evidence.

6.1 Firm Survival and Entry

A natural first margin is whether the increase in the number of firms reflects new business formation, reduced exit, or both. If the truce improved local operating conditions in areas with higher measured pre-truce gang exposure, existing firms may have been better able to remain open, even if new entry adjusted more gradually. Table 2 reports difference-in-differences estimates for business closures and openings. Within each outcome, the *Main* specification uses measured pre-truce gang exposure, while the *Split* and *Top* specifications report complementary exercises based on realized homicide-reduction intensity.

The estimates show that areas with higher measured pre-truce gang exposure experienced both fewer closures and more openings after the truce, relative to areas with lower or unobserved measured exposure. Closures fall by about 10 percent, while openings increase by roughly 8 percent; both estimates are statistically significant. The reduction in closures is especially informative because it provides a direct margin through which the stock of firms can rise: businesses appear more likely to remain active when local security conditions improve. The positive opening estimate suggests that entry also responds, but the survival margin is the clearest and most immediate firm-side

adjustment behind the increase in firm counts.

The complementary homicide-reduction columns show a similar sign pattern across increasingly selective realized-decline classifications. This intensity exercise is not the primary exposure design, but it is useful because it shows that the survival margin remains visible when using a measure tied directly to the magnitude of the observed violence decline.

Table 2: Business Closures and Openings by Measured Gang Exposure

Variables	Closing business					Opening business				
	Main	Split	Top 75	Top 50	Top 25	Main	Split	Top 75	Top 50	Top 25
Estimate	-0.082** (0.036)	-0.041** (0.018)	-0.059*** (0.022)	-0.076** (0.030)	-0.108** (0.051)	0.074** (0.035)	0.035* (0.019)	0.040* (0.023)	0.056* (0.030)	0.065 (0.048)
Per. change	-10.2%	-9.7%	-11.4%	-12.3%	-13.5%	8.1%	5.7%	5.4%	6.4%	6.7%
Mean	0.805	0.422	0.519	0.618	0.799	0.911	0.610	0.738	0.876	0.974
R-squared	0.7826	0.7825	0.7905	0.8056	0.8324	0.8321	0.8320	0.8408	0.8549	0.8777
Obs.	25296	25296	22192	19104	16016	25296	25296	22192	19104	16016
Clusters	1581	1581	1387	1194	1001	1581	1581	1387	1194	1001

*Notes: This table reports difference-in-differences estimates for business closures and openings among small and medium firms (bottom 95%). The first five result columns report closing outcomes; the next five report opening outcomes. Within each outcome, Main uses the main exposure indicator, which equals one for polygons with at least one observed gang-related homicide before the truce and zero for polygons with no such observed homicide. The remaining specifications use complementary classifications based on realized homicide reductions during the truce. Split compares polygons in the top half of the homicide-reduction distribution to polygons in the bottom half. Top 75, Top 50, and Top 25 further restrict the high-decline group within that top half, retaining only polygons above the 75th, 50th, and 25th percentiles of realized homicide reductions among high-decline polygons, respectively. Percentage changes are computed relative to the pre-truce mean reported in the table. All regressions include unit and time fixed effects. Standard errors, clustered at the geographic unit, are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

These patterns also help interpret the main employment and firm-count results. They suggest that the truce did not only coincide with expansion among new firms; it also appears to have reduced the rate at which existing businesses exited the market. This is consistent with lower violence relaxing day-to-day constraints on operating a business, allowing small and medium-sized establishments to remain open longer.

One potential concern is that the increase in registered firms could partly reflect transitions from informal to formal activity, since the administrative data only cover formal-sector firms. This margin remains possible. However, it is unlikely to account for the full pattern. The clearest response is a

decline in closures, which is less naturally explained by new formal registrations alone. Moreover, the World Bank's Doing Business reform history records no business-registration reform in El Salvador in DB2011, DB2012, DB2013, or DB2014; the last recorded reform on this margin occurred in DB2009 (World Bank, 2013). This makes it unlikely that a contemporaneous formalization or registration reform explains the timing of the firm response. Aggregate household-survey evidence from the World Development Indicators provides useful context: employment-to-population rose during the truce period before returning to pre-truce levels (Online Appendix Figure B.5 and Table B.1). Because these aggregate series do not vary by measured exposure, they do not resolve the local formalization concern. They are nevertheless consistent with the interpretation that the truce period coincided with broader net employment gains, not only changes inside the formal-firm registry.

6.2 Sectoral Composition and Local Demand

I next examine whether the response is concentrated in sectors that are more dependent on local market conditions. Following Knight and Johnson (1997), I classify sectors as tradable or non-tradable based on their exposure to broader markets. This comparison is useful because a response dominated by local demand or local operating frictions should appear more strongly in non-tradable activities than in sectors whose demand is less tied to nearby consumers.

Table 3 reports the sectoral decomposition. The pooled sector columns estimate a single stacked regression with polygon-sector and time fixed effects, allowing the post-truce response to differ directly between tradable and non-tradable activity. The estimates show that the positive aggregate effect is concentrated in non-tradable sectors. Tradable activity declines modestly after the truce, by about 2.2 workers and 0.2 firms per polygon, while non-tradable activity increases by about 18.2 workers and 1.4 firms.

Table 3: Difference-in-Differences Estimates by Tradability

Specification	Workers		Number of firms	
	Baseline DID	Pooled sector	Baseline DID	Pooled sector
Baseline DID	16.027*** (5.747)		1.175*** (0.366)	
Tradable		-2.182*** (0.724)		-0.210*** (0.063)
Non-tradable		18.209*** (5.389)		1.384*** (0.334)
Mean	267.543	267.543	38.314	38.314
R-squared	0.9958	0.9956	0.9986	0.9986
Obs.	25296	50592	25296	50592
Clusters	1581	1581	1581	1581

*Notes: This table reports difference-in-differences estimates for workers and the number of firms. The pooled sector columns estimate a stacked sector-level specification with polygon-sector fixed effects and time fixed effects. Tradable and non-tradable sectors are classified following Knight and Johnson (1997). Standard errors, clustered at the geographic unit, are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

The more disaggregated sector patterns point in the same direction. Online Appendix Figure B.8 shows that the largest absolute gains are concentrated in activities closely tied to local commerce and services, especially **G** (*Commerce*), **K** (*Real Estate, Renting, and Business Activities*), and **I** (*Transport, Storage, and Communications*). Other locally oriented activities, such as **H** (*Hotels and Restaurants*), **O** (*Other Community, Social, and Personal Services*), and **F** (*Construction*), also display positive responses, particularly when scaled relative to their smaller pre-truce baselines. This composition strengthens the interpretation that the response is linked to local market activity and day-to-day operating conditions rather than to a broad expansion in externally oriented production.

This sectoral pattern is informative because tradable activities provide a benchmark for broader demand or economy-wide changes. If the results were driven primarily by a general macroeconomic expansion, one would expect more similar responses across tradable and non-tradable activities. Instead, the contrast points toward margins that depend on local operating conditions: nearby consumers, local mobility, the ability of workers and customers to reach firms, and the day-to-day costs of keeping establishments open. These patterns do not identify a single channel, but they make a broad externally oriented investment response less likely to explain the full pattern.

6.3 Wages and Local Labor Market Adjustment

Wages provide a useful, but more limited, way to assess whether the employment gains were accompanied by changes in local labor-market conditions. Safer conditions could increase labor supply by making work less costly, while also increasing labor demand by raising local activity. The wage evidence helps assess which force appears more salient, although the estimates are imprecise. Because wage measures are noisy in small geographic units, I restrict the sample to polygons with more than 10 total employees and code wages as missing when the corresponding worker denominator is zero.

Appendix Table A.6 reports estimates for monthly wages per worker. Under the main measured-exposure specification, the coefficients are positive for the bottom 95%, top 5%, and all-firms samples, but they are not statistically significant. The realized homicide-reduction columns are larger: wages in the bottom 95% increase by about 2.1 percent and wages for all firms increase by about 1.9 percent, with the former statistically significant and the latter marginally significant.

I interpret this evidence cautiously. The wage estimates are less precise than the employment, firm-count, and closure results, and they are not strong enough to establish a separate labor-market channel. At the same time, the estimates do not point to wage declines, which would be more natural if the employment gains reflected only an outward shift in labor supply. They are therefore best read as supportive evidence that the employment gains are compatible with demand-side pressure being an important margin, while leaving substantial uncertainty about the extent of wage adjustment.

6.4 Confidence and Perceived Crime Constraints

I close with broader contextual evidence from FUSADES Economic Outlook Reports. These data are aggregate and are not part of the polygon-level empirical design, but they are useful because they measure perceived constraints and expectations during the truce. Appendix Figure A.7 reports three series: the share of firms identifying different factors as the main obstacle to investment, a consumer confidence index based on households' perceptions of current conditions, major purchase decisions, and economic expectations, and a business confidence index based on firms' assessments of current and expected economic conditions.

The most informative pattern is the timing of consumer confidence. The index rises sharply around the onset of the truce. This timing is consistent with a setting in which improved security made households more willing to move, purchase, and transact locally, making demand-side re-

sponses more plausible. Business confidence does not show a similarly timed change around the onset of the truce, so I place less weight on this series.

The perceived investment constraints point in the same direction from the firm side. The share of firms citing crime as the main obstacle to investment fell from roughly 42 percent before the truce to about 27 percent during the truce, before returning to approximately its pre-truce level afterward. Over the same period, uncertainty became relatively more salient as crime became less frequently cited as the main obstacle. Because these are main-obstacle shares, I interpret this pattern descriptively rather than as a separate test of which constraints changed. I therefore do not treat these aggregate series as mechanism tests. Instead, they provide external context for the firm-level evidence: during the same period in which areas with higher measured pre-truce gang exposure show fewer closures, more openings, and stronger activity in local-facing sectors, consumers became more optimistic and firms reported that crime was a less binding investment obstacle.

Overall, the evidence does not isolate a single mechanism, and several margins likely moved at the same time. The clearest patterns are fewer firm closures and stronger activity in locally oriented sectors. Wage estimates and confidence indicators are less definitive but move in a direction consistent with stronger local activity. Taken together, the evidence suggests that improved local operating conditions and demand-side responses were important margins behind the main effects, even if supply-side adjustment and entry also contributed.

7 Discussion

The mechanism evidence points most clearly to improved firm survival and stronger activity in locally oriented sectors. Still, the reduced-form evidence leaves open interpretive questions that a simple theoretical framework can help organize (formal details in Appendix B.4). First, when safer conditions can increase both labor supply and local demand, what makes the demand-side margin more likely to dominate? Second, when economic activity rises in more-exposed areas, should this be interpreted as net expansion or as spatial reallocation from less affected areas?

The first issue turns on the demand weight of the local market exposed to the improvement in security. This is different from saying simply that larger places have larger effects. In the framework, the relevant object is the potential local demand in the market where fear and transaction costs fall, rather than the local supply of workers per se. When many consumers and firms are exposed to that decline, a given improvement in security can generate a larger consumer-side response: households

are more willing to move, purchase, and interact locally, and firms serving nearby consumers face better operating conditions. This is particularly relevant in El Salvador, where gang activity affected economically meaningful urban and commercial areas, making it plausible that the truce-related decline in violence translated into visible changes in more exposed local markets. In that case, the demand response is more likely to be large relative to the labor-supply response. This is the sense in which the economic weight of the exposed local market matters for interpretation.

This distinction helps read the evidence from the mechanism section. If the decline in violence reduced local fear and transaction costs in economically meaningful local markets, the expected pattern is closer to what appears in the data: fewer firm exits, stronger activity in sectors tied to nearby consumers, and no evidence of wage declines.

The second issue is reallocation. In the framework, a decline in crime increases aggregate employment because safer conditions expand both the willingness to work and the demand for local goods. At the same time, local gains can include spatial reallocation: activity may shift from less affected areas toward areas where security improves. The extent of this reallocation depends on which force dominates. If the demand-side response is strong enough to make the newly safer market relatively more attractive, some workers, customers, or business activity may move toward the more-exposed locations. If the labor-supply response is stronger, aggregate employment can rise with less displacement from other areas.

The evidence cannot fully separate these margins, but it makes a full-reallocation interpretation less compelling. Firm closures fall, which is difficult to explain solely by activity moving across space. The aggregate household-survey evidence also points to a temporary increase in employment-to-population during the truce, consistent with some net expansion in labor-market activity. Taken together, these patterns suggest that some spatial reallocation may have occurred, but the response is unlikely to be only a reshuffling of activity across locations.

This possibility also matters for how exposure boundaries are drawn. If the response operates through local demand and market interactions, the gains need not stop at the polygon boundary used to measure exposure. Nearby areas may share workers, consumers, transport corridors, or commercial links, and improvements in lower-exposure nearby areas would make the baseline contrast conservative. The next section examines this possibility.

8 Additional Evidence on Spillovers

I next examine whether the economic gains documented above extended beyond directly exposed polygons. Spillovers matter for two reasons. First, neighboring areas may also benefit from improved security through shared demand, mobility, commuting, or commercial linkages. Second, indirect exposure can affect interpretation if neighboring polygons that respond to the truce remain in the comparison group. This does not invalidate the measured-exposure comparison, but it clarifies the estimand: the baseline estimates capture differential gains by measured exposure, not the total effect of the truce relative to unaffected areas. A natural concern is that a narrow exposure definition may leave indirectly affected neighboring polygons in the comparison group. If lower-exposure or neighboring areas also benefit from improved security, the baseline contrast would be consistent with broader local gains rather than isolate a pure direct effect.

To assess this possibility, I augment the baseline specification with measures of neighboring exposure. The first measure indicates whether a polygon has spatial contact with a polygon that had measured pre-truce gang exposure. The second measure uses proximity to the nearest exposed polygon, with higher values corresponding to closer locations. These exercises should be interpreted cautiously. The exposure proxy is measured with error, and spatial contact or proximity captures only one form of interference; spillovers may also operate through commuting, commercial linkages, transport routes, firm relocation, or broader market connections.

Appendix Table A.7 reports the corresponding estimates. The direct-exposure coefficient remains positive after accounting for neighboring exposure. The neighbor and proximity-based measures are also positive, suggesting that economic gains may have extended beyond directly exposed polygons. This pattern is consistent with local improvements spreading through nearby markets, although the estimates are best read as suggestive evidence rather than a separate source of identification.

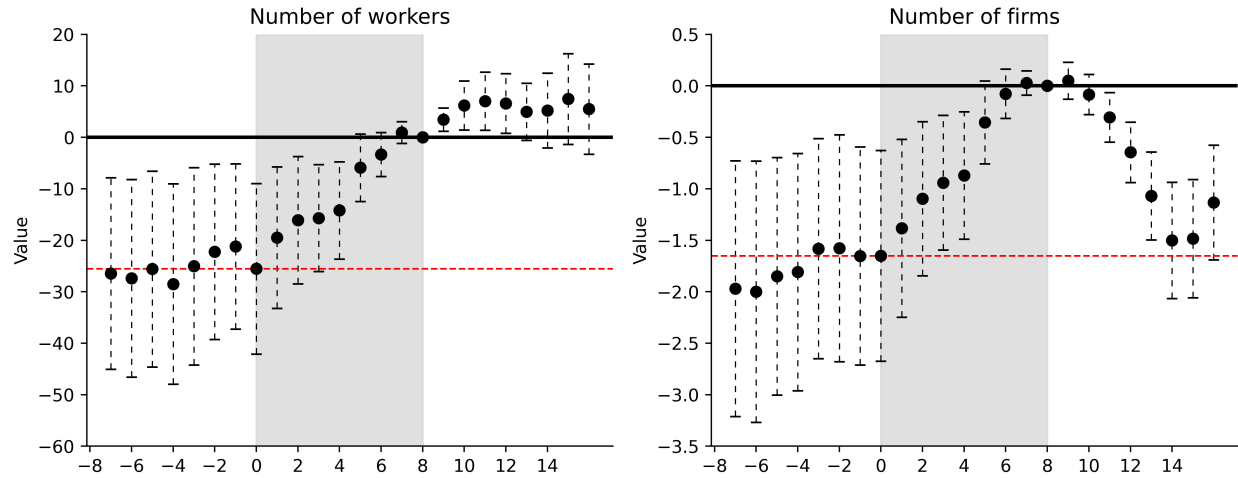
As an additional check motivated by the concern that exposure boundaries may be too narrow, Appendix Table A.8 groups directly exposed polygons and spatial-contact polygons into a single exposed category. This exercise asks whether the main pattern survives when the exposure boundary is drawn more broadly. The estimates remain positive for both employment and firm counts, suggesting that the main findings are robust to this broader exposure definition, and are consistent with economic gains extending to neighboring polygons.

9 Aftermath of the Truce: Rising Crime and Economic Reversal

Shortly after the truce dissolved, the national homicide rate began to rise steadily, eventually surpassing its pre-truce level. Appendix Figure A.8 shows this reversal. By 2015, El Salvador recorded the most violent year in its recent history, with the national homicide rate exceeding 100 per 100,000 inhabitants—implying that nearly 0.1% of the population was killed that year. This surge in violence coincided with a sharp escalation in the government’s anti-gang efforts, often referred to as *mano dura* (“iron-fist”) policies, which emphasized aggressive policing. These confrontational tactics intensified conflict between security forces and gang networks, likely fueling further instability and reversing many of the temporary security gains achieved during the truce.

In tandem with rising crime, the economic indicators that had improved during the truce began to reverse. Figure 4 shows that while the growth in employment levels decelerated, it remained above the pre-truce baseline for most other periods. Similar to the pre-truce period, employment began to follow the same trajectory as its counterfactual, suggesting that once violence resurged, labor market dynamics reverted to their previous patterns. By contrast, the total number of firms steadily declined, ultimately returning to its pre-truce levels by the end of 2015. Interestingly, both higher- and lower-exposure areas experienced increases in crime after the ceasefire collapsed, yet higher-exposure areas exhibited a sharper decline in active firm counts as violence rose. One possible explanation is that these areas had shown larger gains during the truce—consistent with a substantial “peace dividend”—and were therefore more vulnerable to resurgent insecurity. This suggests that firms in higher-exposure areas may have adjusted their operations under the assumption of sustained improvements in security, making them more susceptible to instability when violence returned.

Figure 4: Trends in the Number of Workers and Firms Post-Truce



Notes: This figure presents event-study estimates from a specification analogous to Equation (1), redefining the event time to center around the end of the gang truce. The estimates compare areas with higher measured pre-truce gang exposure to areas with lower or unobserved measured exposure. The dependent variables are the number of workers (left panel) and number of firms (right panel). The red dotted horizontal line indicates the estimated pre-truce reference level, which serves as a benchmark for assessing how much of the truce-period gain remains after the truce ends. The shaded gray area highlights the duration of the truce. Time is grouped into three-month intervals. All regressions include unit and time fixed effects. Vertical bars represent 95% confidence intervals, with standard errors clustered at the geographic unit level.

This reversal should be interpreted descriptively, since the end of the truce coincided with broader policy changes, including the escalation of *mano dura* strategies. Still, the timing and composition of the reversal are informative. Areas with higher measured exposure, which had benefited more during the truce, also show a sharper decline in the number of firms once violence re-escalated. This pattern is consistent with the idea that smaller or more vulnerable businesses expanded or survived under safer conditions but became more exposed when insecurity returned. Employment appears more resilient: although job growth slowed after the truce, it did not fall below pre-truce levels and eventually returned to its earlier trajectory.

Additional context comes from a later, more limited decline in homicides in early 2016, following a non-aggression pact among gangs widely understood as a response to the government's intensifying security measures (Gagne, 2017). Although the agreement lowered the national homicide rate, crime levels remained close to their pre-truce average and public sentiment did not improve: consumer confidence and perceptions of safety showed no meaningful gains (see Online Appendix

Figure B.11). This contrast may help explain why Brown et al. (2024), studying firm outcomes during that period using data from a large national distributor, finds no positive effects. That pattern is consistent with the evidence in this paper that large firms and tradable sectors were less responsive to improvements in local security.

10 Conclusion

This paper studies how a sharp reduction in homicides during the 2012 Salvadoran gang truce relates to formal employment and registered firm activity. The truce produced a large decline in homicides, while areas differed in their measured pre-truce exposure to gang activity. I use this variation to compare post-truce changes in areas with higher measured gang exposure to changes in areas with lower or unobserved measured exposure.

The results show meaningful gains in local economic activity following the truce. Areas with higher measured pre-truce gang exposure experienced larger post-truce increases in employment and in the number of firms, with the strongest and most stable responses concentrated among small and medium-sized establishments. The estimates remain positive under matched comparisons that improve balance in pre-truce economic scale and under a synthetic difference-in-differences exercise that improves pre-truce fit. A complementary specification based on realized homicide reductions also shows a positive gradient: larger declines in violence are associated with larger improvements in employment and firm activity.

The evidence on margins of adjustment is most consistent with firm survival and stronger activity in locally oriented sectors playing an important role. Areas with higher measured exposure experienced fewer firm closures and more openings, with the decline in closures providing a particularly clear margin for the increase in firm counts. Sectoral results show that the gains are concentrated in non-tradable and local-facing activities, including commerce, services, transport, hotels and restaurants, and construction. Wage estimates and aggregate consumer confidence indicators are less definitive, but they move in a direction consistent with improved local operating conditions and stronger local demand.

The analysis also suggests that these gains may have extended beyond narrowly measured exposure boundaries. Neighboring areas show suggestive evidence of improved outcomes, and specifications that group polygons with measured pre-truce exposure and adjacent polygons produce similar positive patterns. At the same time, the gains appear fragile. When violence rose again after the

collapse of the truce, firm-count improvements reversed more clearly than employment, suggesting that the businesses most responsive to safer conditions were also vulnerable to renewed insecurity.

Taken together, the findings indicate that large reductions in violence can be followed by meaningful improvements in formal employment and firm activity, especially among smaller and more locally embedded firms. The evidence highlights the economic relevance of local security conditions, while also underscoring that these gains depend on the durability of violence reductions.

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A Appendix

A.1 Figures and Tables

Table A.1: Summary Statistics by Pre-Truce Gang Exposure

Variable	Lower measured exposure		Higher measured exposure	
	Mean	SD	Mean	SD
<i>Panel A</i>				
Number of firms per polygon	9.81	29.39	38.59	122.91
Number of firms per polygon (excl. top 5%)	9.28	26.96	36.32	113.97
Number of workers per polygon	168.41	736.72	735.71	2773.39
Number of workers per polygon (excl. top 5%)	63.95	219.33	267.91	942.32
Average wages	321.72	227.38	356.73	211.91
Average wages (excl. top 5%)	276.58	103.31	302.12	98.15
Number of opened firms per quarter	0.30	1.22	0.99	3.72
Number of closed firms per quarter	0.18	0.68	0.79	2.46
<i>Panel B</i>				
Homicide rate (per 100k inhabitants) before truce	6.73	19.72	7.48	13.19
Homicide rate (per 100k inhabitants) after truce	4.14	16.68	3.92	10.36
Number of polygons	1283.00		298.00	

Notes: Panel A reports descriptive statistics on business activity, while Panel B reports descriptive statistics on violence before and after the truce. Areas are classified using the pre-truce gang-exposure proxy based on whether a polygon recorded at least one gang-related homicide before the truce.

Table A.2: Summary Statistics: Treatment and $NN(5)$ Matched Control Areas

Variable	Matched Control		Treatment	
	Mean	SD	Mean	SD
<i>Panel A</i>				
Number of firms per polygon	8.65	16.43	11.57	15.66
Number of firms per polygon (excl. top 5%)	8.28	15.59	11.12	15.14
Number of workers per polygon	128.99	391.15	158.60	324.87
Number of workers per polygon (excl. top 5%)	53.31	111.16	71.59	111.08
Average wages	314.27	192.49	312.69	149.48
Average wages (excl. top 5%)	274.18	102.04	281.34	90.20
Number of opened firms per quarter	0.26	0.83	0.34	0.73
Number of closed firms per quarter	0.16	0.55	0.27	0.61
<i>Panel B</i>				
Homicide rate (per 100k inhabitants) before truce	6.99	19.95	7.62	13.85
Homicide rate (per 100k inhabitants) after truce	4.23	16.88	4.00	11.02
Number of polygons	1065.00		213.00	

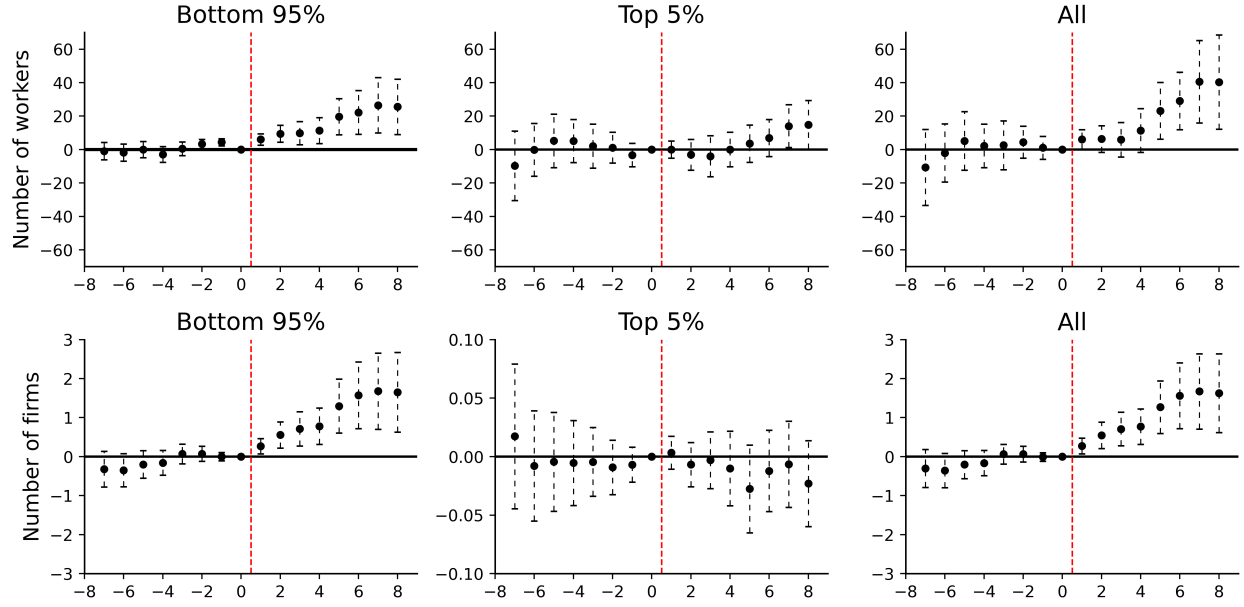
Notes: Panel A reports descriptive statistics on business activity, while Panel B reports descriptive statistics on violence before and after the truce. $NN(k)$ denotes k -nearest-neighbor matching on pre-truce firm counts. The matched-control group corresponds to the preferred $NN(5)$ specification, which uses a caliper and matches without replacement.

Table A.3: Difference-in-Differences Estimates: Baseline, SDID, and $NN(5)$ Matched Specifications

Variables	Workers			Number of firms		
	Baseline	SDID	Matched	Baseline	SDID	Matched
Treatment	16.027*** (5.747)	9.120* (5.090)	2.333* (1.341)	1.175*** (0.366)	0.960*** (0.320)	0.467*** (0.155)
Mean	267.543	267.543	71.368	38.314	38.314	11.627
R-squared	0.9958		0.9867	0.9986		0.9933
Obs.	25296	25296	20448	25296	25296	20448
Clusters	1581		1278	1581		1278

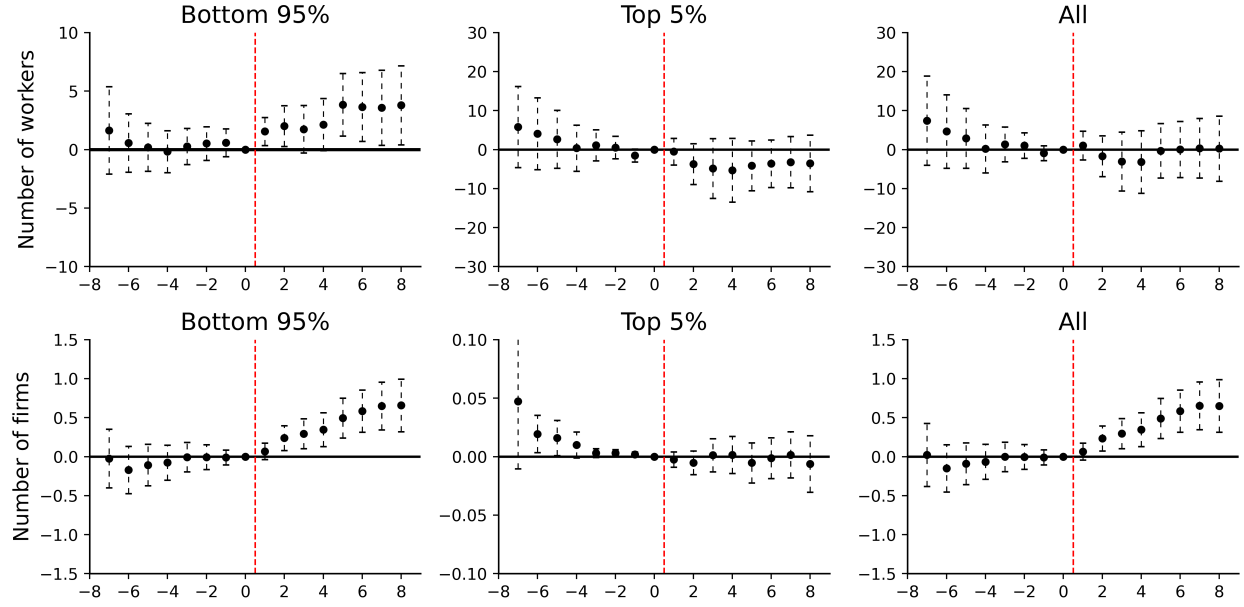
Notes: This table reports difference-in-differences estimates for the effect of the truce on the number of workers and firms. The baseline columns use the full set of lower-exposure polygons as the comparison group, while the SDID and matched columns provide alternative counterfactual constructions based on synthetic difference-in-differences and the preferred $NN(5)$ matched sample. All regressions include unit and time fixed effects. Standard errors are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure A.1: Event-Study Estimates for Workers and Firms, Full Sample



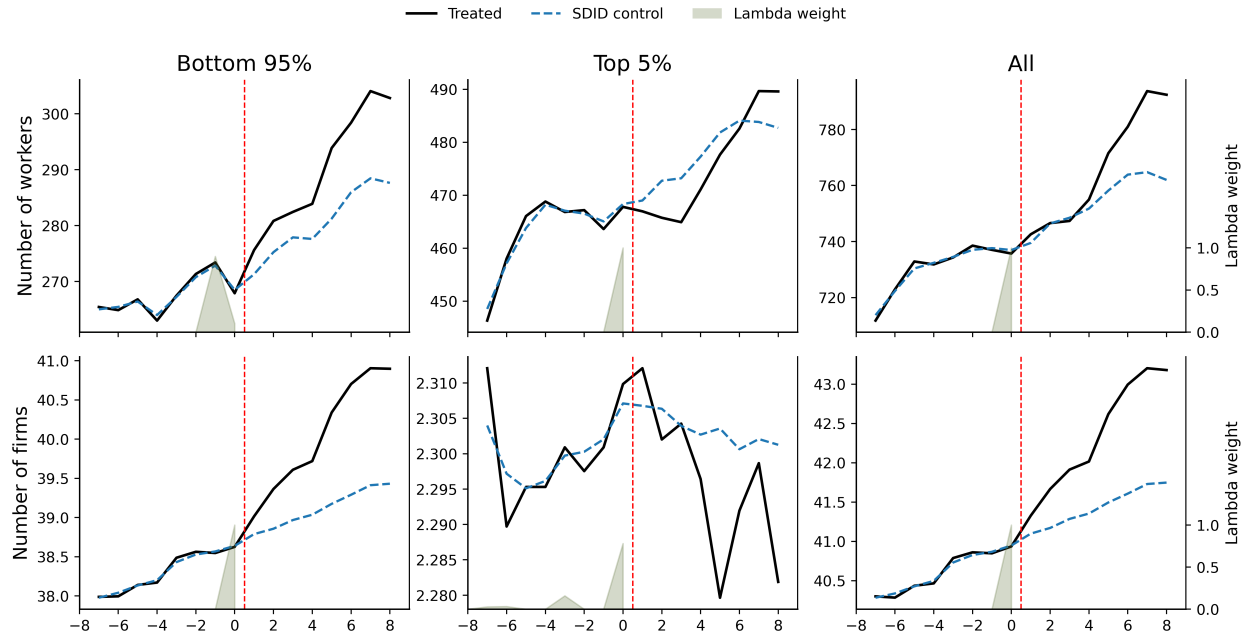
Notes: This figure displays event-study estimates from Equation (1) for the full sample of firms. The X-axis consolidates data into three-month intervals, spanning 24 months before and after the truce. All regressions include unit and time fixed effects. Vertical bars indicate 95% confidence intervals, with standard errors clustered at the geographic unit level.

Figure A.2: Event-Study Estimates Using $NN(5)$ Matched Controls



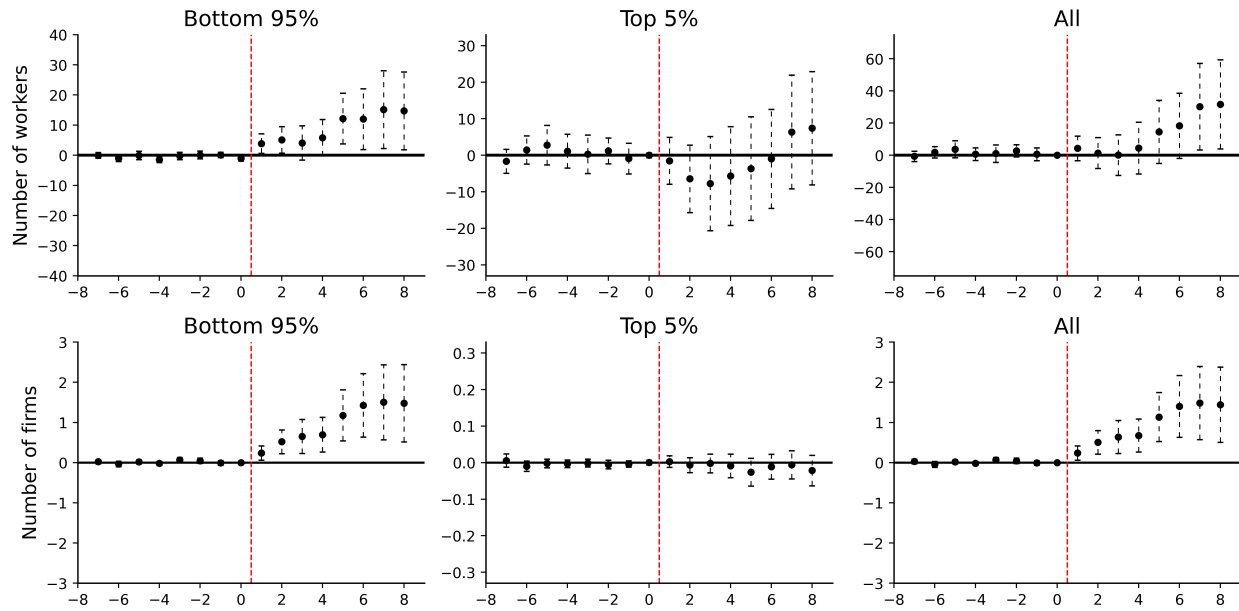
Notes: This figure displays event-study estimates from Equation (1) using the matched-control design. The X-axis consolidates data into three-month intervals, spanning 24 months before and after the truce. All regressions include unit and time fixed effects. Vertical bars indicate 95% confidence intervals, with standard errors clustered at the geographic unit level. The comparison group is restricted to the preferred $NN(5)$ matched controls.

Figure A.3: Synthetic Difference-in-Differences Trajectories



Notes: This figure presents synthetic difference-in-differences trajectories for the main specification based on pre-truce gang exposure. The figure compares the evolution of outcomes in areas with higher measured gang exposure to the synthetic comparison trajectory constructed from areas with lower or unobserved measured exposure.

Figure A.4: Synthetic Difference-in-Differences Event-Study Estimates



Notes: This figure presents event-study estimates from the synthetic difference-in-differences robustness exercise for the main specification based on pre-truce gang exposure. The estimates provide an alternative counterfactual construction for comparing areas with higher measured gang exposure to areas with lower or unobserved measured exposure around the truce. Confidence intervals are constructed using 100 bootstrap replications.

Table A.4: Continuous Difference-in-Differences Estimates Based on Homicide Reductions

Dependent	Model	One Year		Two Years	
		Coef/SE	Per. Change	Coef/SE	Per. Change
Workers	Linear	12.72***	(8.40%)	26.11***	(17.24%)
		(3.71)		(7.16)	
	Quadratic	12.90**	(8.52%)	26.46**	(17.48%)
		(6.49)		(12.50)	
Firms	Linear	1.01***	(4.55%)	1.93***	(8.68%)
		(0.26)		(0.47)	
	Quadratic	1.03**	(4.61%)	1.95**	(8.78%)
		(0.46)		(0.81)	

Notes: This table reports estimates from the complementary continuous specification based on the magnitude of homicide reductions during the truce. The table reports linear and quadratic specifications, with magnitudes summarized one and two years after the onset of the truce. Percentage changes are computed relative to the relevant pre-truce mean. Standard errors are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

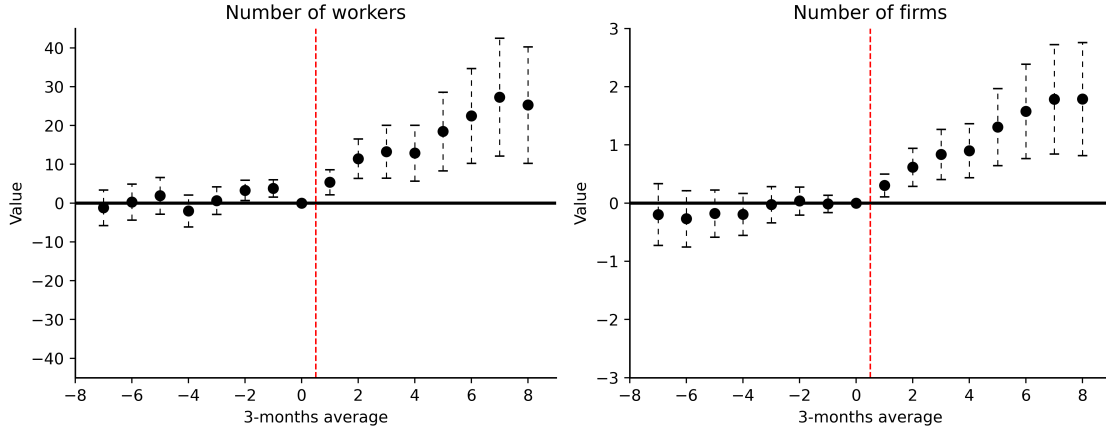
Table A.5: Predictors of Exposure to Homicide Reductions During the Truce

	Exposure to Homicide Reduction					
	(1)	(2)	(3)	(4)	(5)	(6)
Log Avg. Firms (Pre)	0.095* (0.055)	0.050 (0.054)	—	—	0.050 (0.055)	0.017 (0.057)
Log Avg. Workers (Pre)	0.038 (0.038)	0.008 (0.036)	—	—	0.008 (0.036)	-0.022 (0.038)
Average Hourly Wage (Pre)	-0.086** (0.044)	-0.070* (0.040)	—	—	-0.067 (0.042)	-0.045 (0.045)
Firm Growth (Pre)	—	—	0.490 (3.271)	1.231 (2.972)	1.913 (3.047)	-1.402 (3.189)
Worker Growth (Pre)	—	—	1.573 (1.286)	1.091 (1.183)	0.557 (1.220)	1.631 (1.333)
Wage Growth (Pre)	—	—	-1.222 (2.254)	0.205 (2.127)	-0.728 (2.216)	-1.584 (2.707)
Log Population	—	0.269*** (0.040)	—	0.309*** (0.036)	0.269*** (0.040)	0.017 (0.046)
Average Homicide (Pre)	—	0.036*** (0.004)	—	0.037*** (0.004)	0.036*** (0.004)	0.015*** (0.004)
Wage Missing	-0.094 (0.107)	0.022 (0.103)	-0.245*** (0.067)	0.068 (0.074)	0.029 (0.104)	-0.133 (0.104)
R-squared	0.035	0.122	0.011	0.118	0.122	0.464
Obs.	1581	1581	1581	1581	1581	1581

Notes: This table examines whether pre-truce economic conditions predict exposure to homicide reductions during the truce. The dependent variable is exposure to homicide reduction. Economic variables are measured before the truce. Specifications also include population and pre-truce homicide controls where indicated. Standard errors are shown in parentheses.

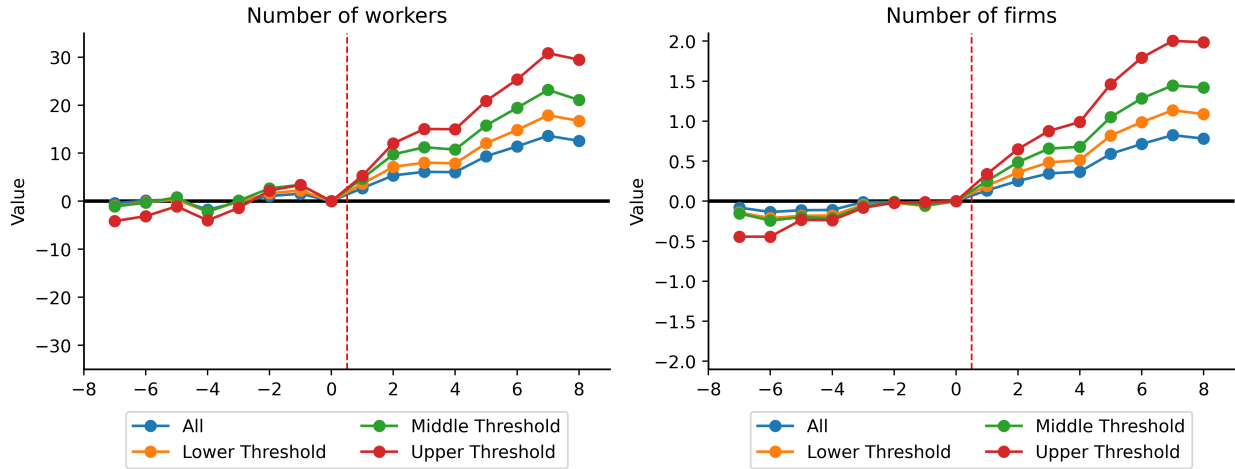
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure A.5: Event-Study Estimates for the Continuous-Treatment Specification



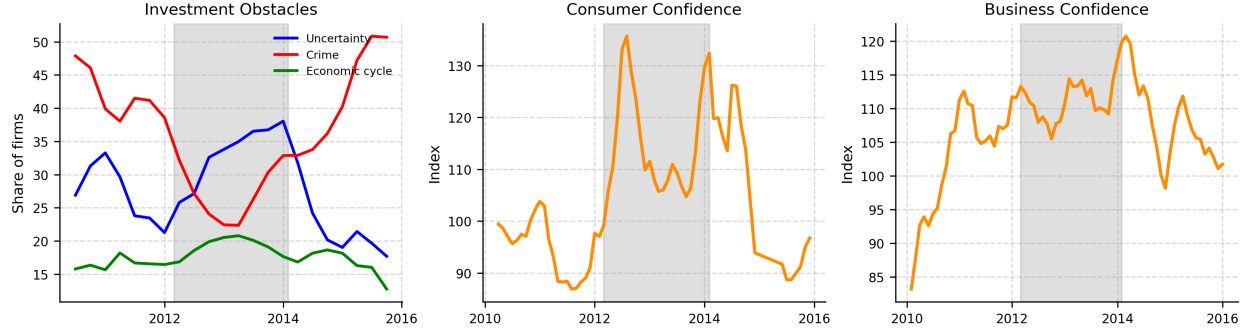
Notes: This figure presents event-study estimates under the complementary continuous-treatment specification, following the linear specification in Equation (4). Treatment intensity is defined as the standardized reduction in homicides during the truce period. The model includes unit and time fixed effects. Vertical bars denote 95% confidence intervals with standard errors clustered at the geographic unit level.

Figure A.6: Event-Study Estimates by Treatment Intensity, Excluding the Top 5% Largest Firms



Notes: This figure presents estimated coefficients from Equation (1), applied separately to groups of polygons classified by the intensity of total homicide reductions during the truce period. The three lines correspond to Low, Intermediate, and High Threshold groups, defined respectively as polygons at or above the 25th, 50th, and 75th percentiles of the homicide-reduction distribution among treated polygons. The X-axis consolidates time into three-month intervals, spanning 24 months before and after the onset of the truce.

Figure A.7: Investment Obstacles and Confidence During the Truce



Notes: The figure plots three aggregate series from FUSADES Economic Outlook Reports: the share of firms citing crime, uncertainty, or the economic cycle as the main obstacle to investment; the consumer confidence index; and the business confidence index. Confidence data are sourced from Economic Outlook Reports published by FUSADES (2010–2020). The shaded area indicates the truce period. These series are used as aggregate contextual evidence and are not part of the main polygon-level design.

Table A.6: Difference-in-Differences Estimates for Wages

Variables	Main exposure			Realized homicide reduction		
	Bottom 95%	Top 5%	All firms	Bottom 95%	Top 5%	All firms
Treatment	1.396 (3.135)	4.995 (9.621)	3.928 (3.825)	6.169** (3.014)	9.356 (9.966)	7.135* (3.873)
Mean	307.896	514.111	366.123	298.541	532.284	372.752
R-squared	0.8827	0.9714	0.9752	0.8829	0.9715	0.9753
Obs.	11676	5221	11808	11676	5221	11808
Clusters	734	335	738	734	335	738

Notes: This table reports difference-in-differences estimates for monthly wages per worker. The main exposure indicator equals one for polygons with at least one observed gang-related homicide before the truce and zero for polygons with no such observed homicide. The second set of columns uses the realized homicide-reduction classification. The sample is restricted to polygons with more than 10 total employees. Wage outcomes are coded as missing when the corresponding worker denominator is zero, so the number of observations varies across firm-size categories. All regressions include unit and time fixed effects. Standard errors, clustered at the geographic unit, are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.7: Spillover Estimates Using Measured Pre-Truce Gang Exposure

Variables	Workers			Number of firms		
	Base	Neighbor	Proximity	Base	Neighbor	Proximity
Treatment	16.027*** (5.747)	19.770*** (5.716)	14.373** (5.793)	1.175*** (0.366)	1.498*** (0.364)	1.031*** (0.373)
Spillover - neighbor		9.079*** (1.776)			0.783*** (0.167)	
Spillover - proximity			0.353*** (0.079)			0.031*** (0.008)
Mean	267.543	267.543	267.543	38.314	38.314	38.314
R-squared	0.996	0.996	0.996	0.999	0.999	0.999
Obs.	25296	25296	25296	25296	25296	25296
Clusters	1581	1581	1581	1581	1581	1581

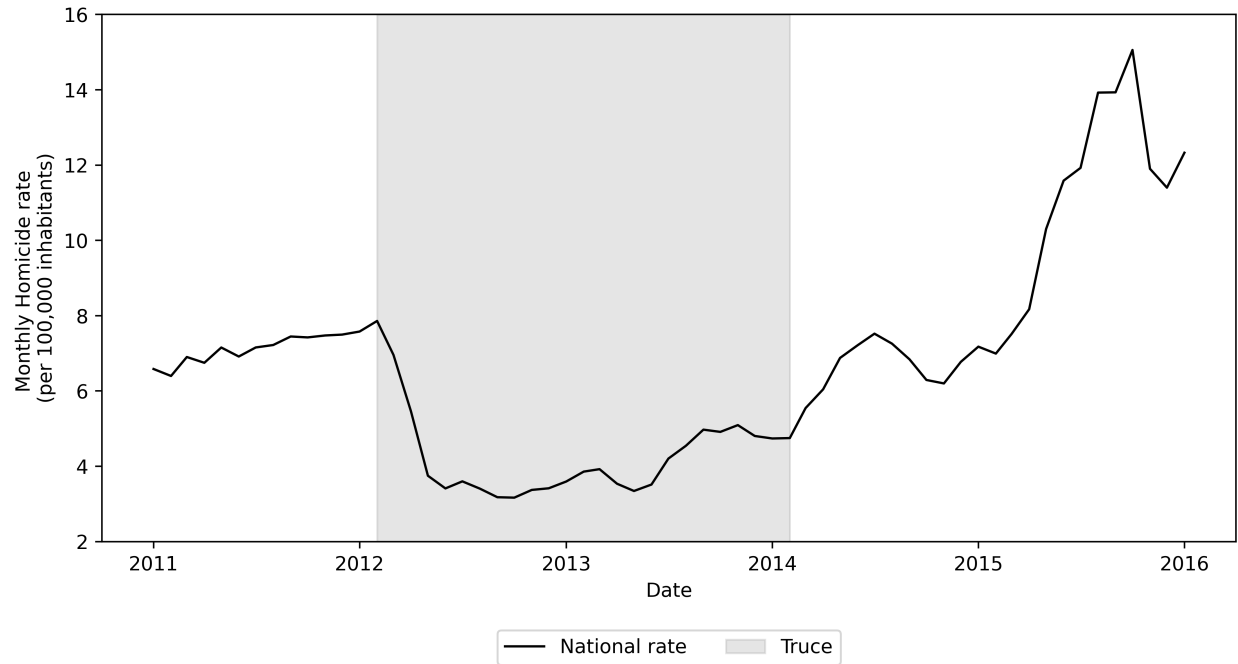
Notes: This table reports difference-in-differences estimates that include direct exposure and neighboring exposure measures. The direct exposure indicator equals one for polygons with at least one observed gang-related homicide before the truce. Spillover – neighbor equals one for polygons with spatial contact with at least one directly exposed polygon. Spillover – proximity is increasing in closeness to the nearest directly exposed polygon. All regressions include unit and time fixed effects. Standard errors are clustered at the geographic unit. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A.8: Estimates Grouping Directly Exposed and Neighboring Areas

Variables	Workers	Number of firms
Treatment + Neighbor	12.931*** (2.382)	1.041*** (0.174)
Mean	171.814	24.873
R-squared	0.996	0.999
Obs.	25296	25296
Clusters	1581	1581

Notes: This table reports difference-in-differences estimates from a specification that redefines exposure to include both polygons with measured pre-truce gang exposure and polygons with spatial contact with exposed polygons. This grouped exposure definition addresses the concern that neighboring areas may be indirectly exposed to the truce-related improvement in security. The mean is computed for the combined directly exposed and neighboring group before the truce. All regressions include unit and time fixed effects. Standard errors are clustered at the geographic unit. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure A.8: National Homicide Rate Around and After the Truce



Notes: This figure plots the national monthly homicide rate per 100,000 inhabitants. The series is smoothed using a trailing 3-month moving average. The shaded area highlights the duration of the truce.

B Online Appendix

B.1 Additional Specifications

B.1.1 Continuous Treatment Specification

This subsection provides additional details on the continuous-treatment specification used as a complement to the baseline binary exposure design in the main text. While the main empirical strategy relies on a proxy for pre-truce gang exposure, this complementary exercise examines whether areas experiencing larger realized declines in violence during the truce also exhibit stronger economic responses.

A key concern in this setting is that, when treatment intensity varies continuously, standard two-way fixed effects (TWFE) estimators may be difficult to interpret in the presence of heterogeneous or nonlinear effects (Callaway, Goodman-Bacon and Sant’Anna, 2024). For this reason, I implement a residualization approach that separates the variation used for identification from fixed differences across polygons and common aggregate shocks over time.

Let $D_i \in \mathbb{R}_+ \cup \{0\}$ denote treatment intensity, where $D_i = 0$ for polygons with no realized homicide decline relative to the reference cutoff, and $D_i > 0$ indexes the magnitude of the observed decline during the truce. In a first step, I estimate the following model using untreated observations:

$$Y_{it} = \alpha_i + \lambda_t + \epsilon_{it} \quad (2)$$

where Y_{it} denotes the outcome of interest for polygon i at time t , α_i are polygon fixed effects, and λ_t are time fixed effects. I then construct residualized outcomes by removing these estimated fixed effects from the observed data:

$$\tilde{Y}_{it} = Y_{it} - \hat{\alpha}_i - \hat{\lambda}_t \quad (3)$$

In a second step, I relate these residualized outcomes to treatment intensity using a flexible dose-response specification. In the empirical implementation, this relationship is estimated using linear and quadratic specifications of treatment intensity, allowing the response of outcomes to vary with the magnitude of the homicide decline.

Let $\Delta y_t(d)$ denote the residualized outcome response at event time t for treatment intensity d . I estimate two parametric dose-response functions. The linear specification is

$$\Delta y_t(d) = \beta_1 d, \quad (4)$$

while the quadratic specification is

$$\Delta y_t(d) = \beta_1 d + \beta_2 d^2. \quad (5)$$

In the linear specification, the marginal response to treatment intensity is constant and given by β_1 . In the quadratic specification, the marginal response varies with intensity and is given by $\beta_1 + 2\beta_2 d$.

This continuous-treatment analysis is interpreted as complementary evidence rather than as the paper's primary identification strategy. It is useful for characterizing how economic outcomes covary with larger observed reductions in violence, but it relies on stronger assumptions than the baseline binary specification and is therefore interpreted more cautiously.

B.2 Construction of the Homicide Reduction Index

To systematically classify treatment and control areas, I construct a standardized measure of abnormal homicide declines. Let $H_{p,t}$ denote the homicide rate in polygon p at month t . I first compute a short-run change in the homicide rate as follows:

$$\Delta H_{p,t} = \left(\frac{1}{36} \sum_{\tau=t-36}^{t-1} H_{p,\tau} \right) - \left(\frac{1}{3} \sum_{\tau=t}^{t+2} H_{p,\tau} \right).$$

This measure $\Delta H_{p,t}$ compares the mean homicide rate over the 36 months prior to t with the mean rate after the event window. In the empirical implementation, I omit the immediate transition months around the onset of the truce before measuring the post-event decline. During the truce, I expect some polygons to exhibit larger values of $\Delta H_{p,t}$, indicating a sharp reduction in total homicide rates.

Next, I standardize each observed drop against the distribution of comparable homicide changes within the same polygon. Define the polygon-specific mean and standard deviation of these changes:

$$\mu_p = \mathbb{E}[\Delta H_{p,t'}], \quad \sigma_p = \sqrt{\mathbb{E}[(\Delta H_{p,t'} - \mu_p)^2]},$$

Using these statistics, I construct a standardized measure:

$$Z_{p,t} = \frac{\Delta H_{p,t} - \mu_p}{\sigma_p}.$$

This $Z_{p,t}$ score indicates how unusual the homicide reduction is for polygon p at time t , relative to its own historical variability. A more positive $Z_{p,t}$ means that the current drop in homicides is large relative to what would typically be expected in that polygon based on past fluctuations.

B.2.1 Placebo tests

The primary modification to the conventional method involves merging multiple data collections, where treatment units are identified based on experiencing an abnormal drop in homicide activity during the truce period, and placebo periods 3, 5, 7, and 9 months prior to the truce. These datasets are then combined, incorporating a categorical variable s , which indicates whether the assignment to treatment or control occurred during the truce. This variable interacts fully with the unit-fixed effects, time-fixed effects, and the treatment indicator.

The modified TWFE model, incorporating the stacked placebo and truce-period datasets and the interaction with a dummy variable for the truce period, is specified as follows:

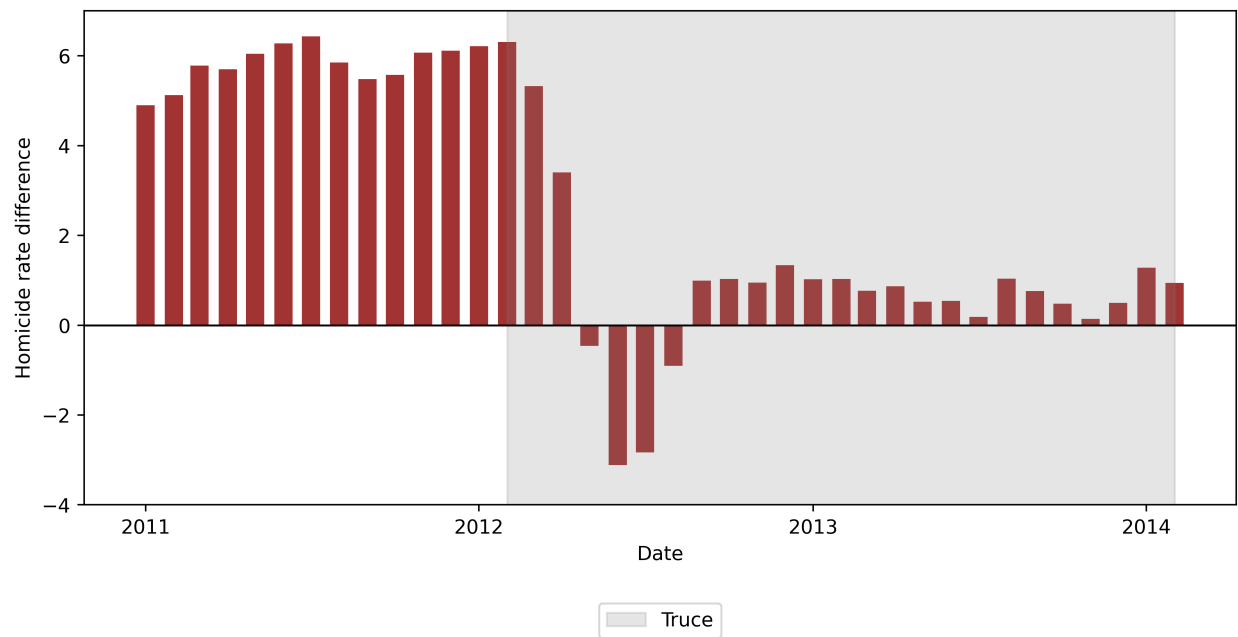
$$Y_{ist} = \alpha + \tau_0 D_{ist} + \tau_1 D_{ist} \mathbb{1}\{s \in Truce\} + \gamma_{is} + \lambda_{ts} + \epsilon_{ist} \quad (6)$$

where Y_{ist} represents the outcome of interest for unit i selected as control or treatment in period s at time t before the event s . D_{ist} is the treatment indicator, activated post-event s for units identified as treated, γ_{is} are the unit fixed effects (polygons) interacted s , λ_{ts} are the time fixed effects interacted s , and ϵ_{ist} is the idiosyncratic error term. The errors are clustered at the level of geographic units interacted with s . The coefficient τ_0 captures the average treatment effect in the placebo scenarios, that is, across periods s preceding the truce when treatment and control units are identified outside the actual intervention window. The coefficient τ_1 measures the incremental effect when the treatment and control selections coincide with the truce period. Hence, the total treatment effect during the real truce is given by the sum $\tau_0 + \tau_1$.

To further dissect the dynamics of the treatment effect, I implement an Event Study analysis. This approach examines the evolution of the impact pre- and post-treatment. The model is augmented with interaction terms between the treatment and time dummies.

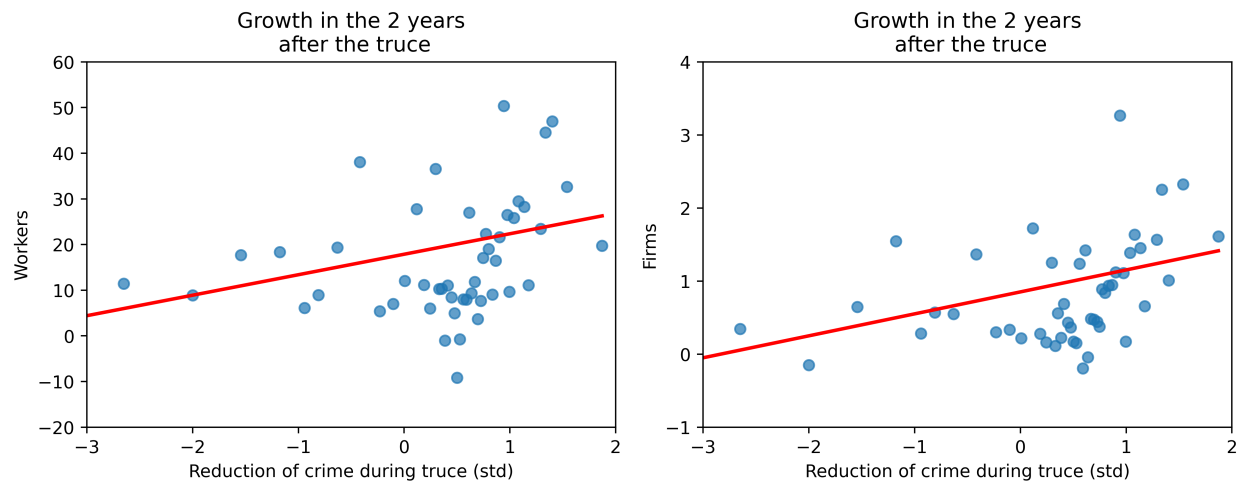
B.3 Figures and Tables

Figure B.1: Difference in Homicide Rates by Realized Homicide Reductions



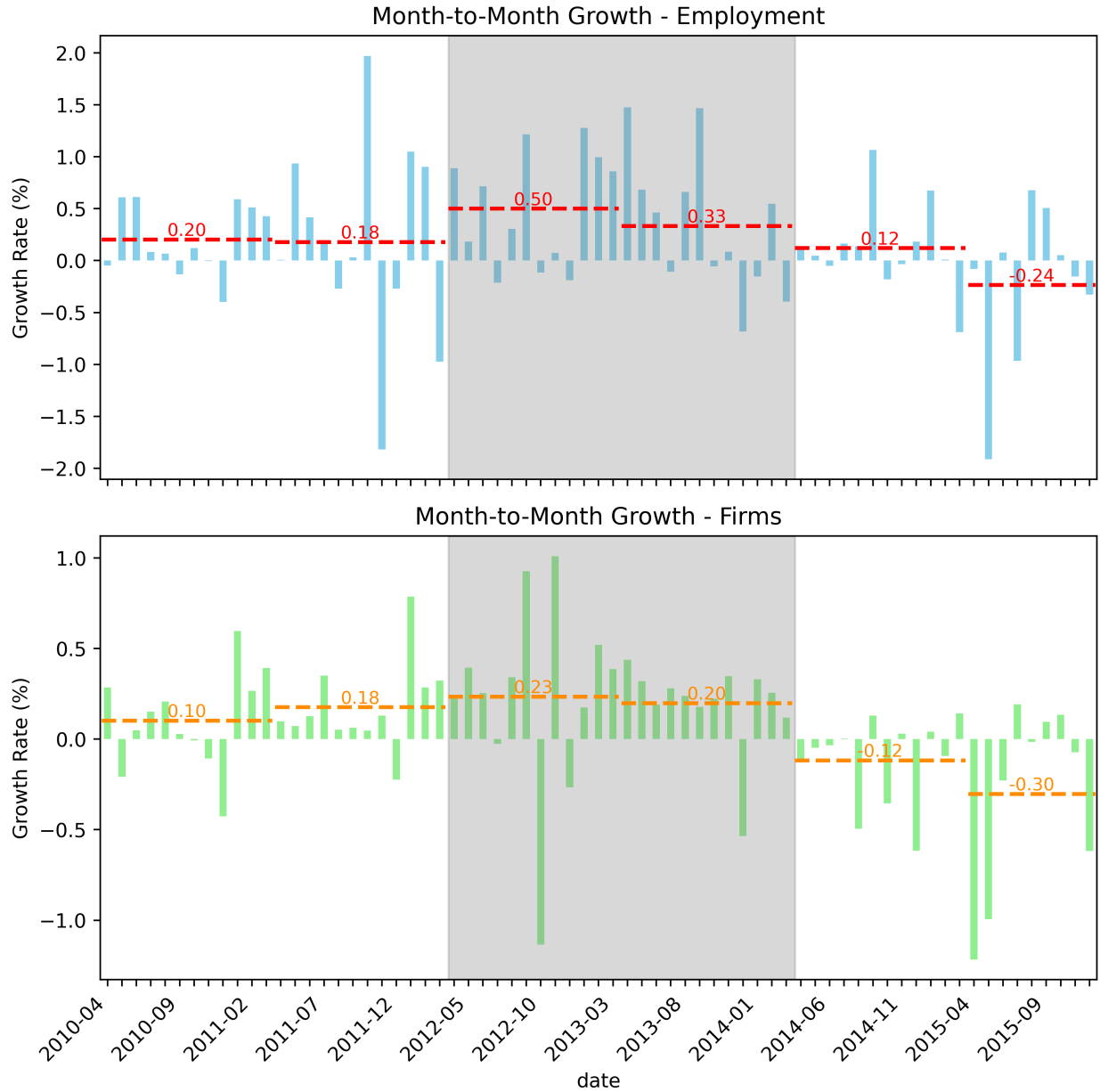
Note: This figure plots the monthly difference in homicide rates between areas classified as treated and control based on realized homicide reductions during the truce. The underlying homicide-rate series are smoothed using a trailing 3-month moving average. The bars summarize how the homicide-rate gap evolves around the onset of the truce.

Figure B.2: Relationship Between Crime Reduction During the Truce and Economic Activity



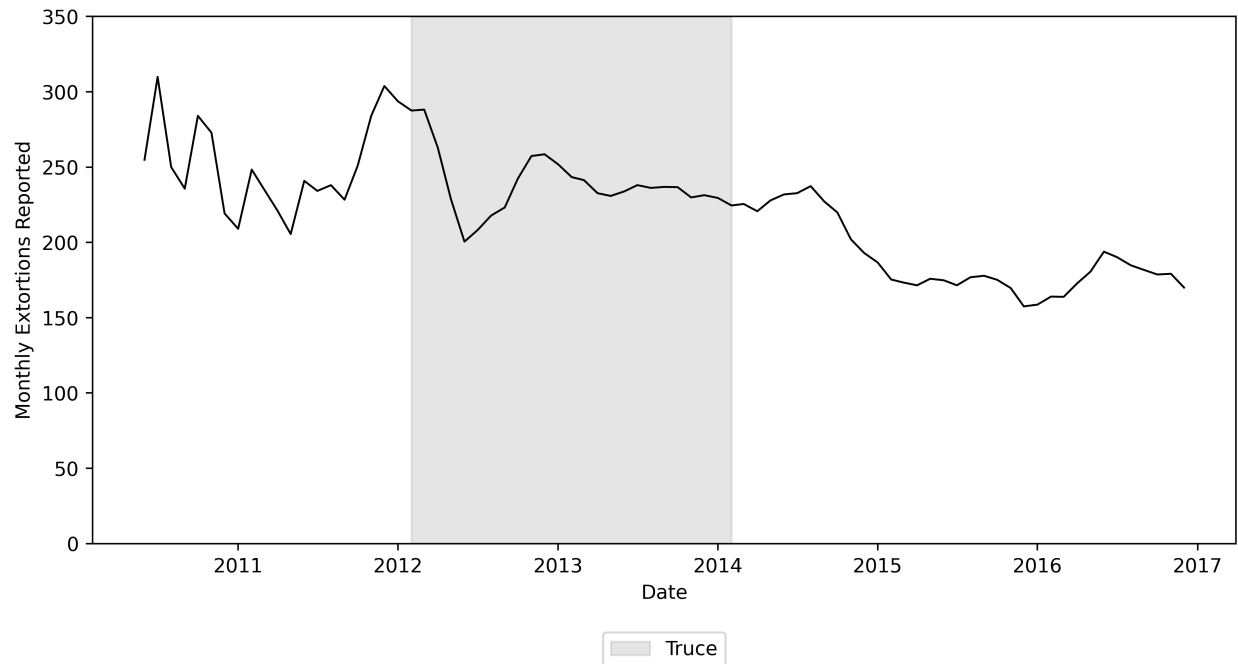
Notes: This figure presents the relationship between the reduction in crime during the truce and two key economic indicators: employment (left panel) and firm presence (right panel). Each point represents a bin of 50 observations, with a fitted linear regression line shown in red. The left panel illustrates the variation in the number of workers, while the right panel depicts the variation in the number of firms, both in response to crime reduction.

Figure B.3: Monthly Growth in Employment and Firms



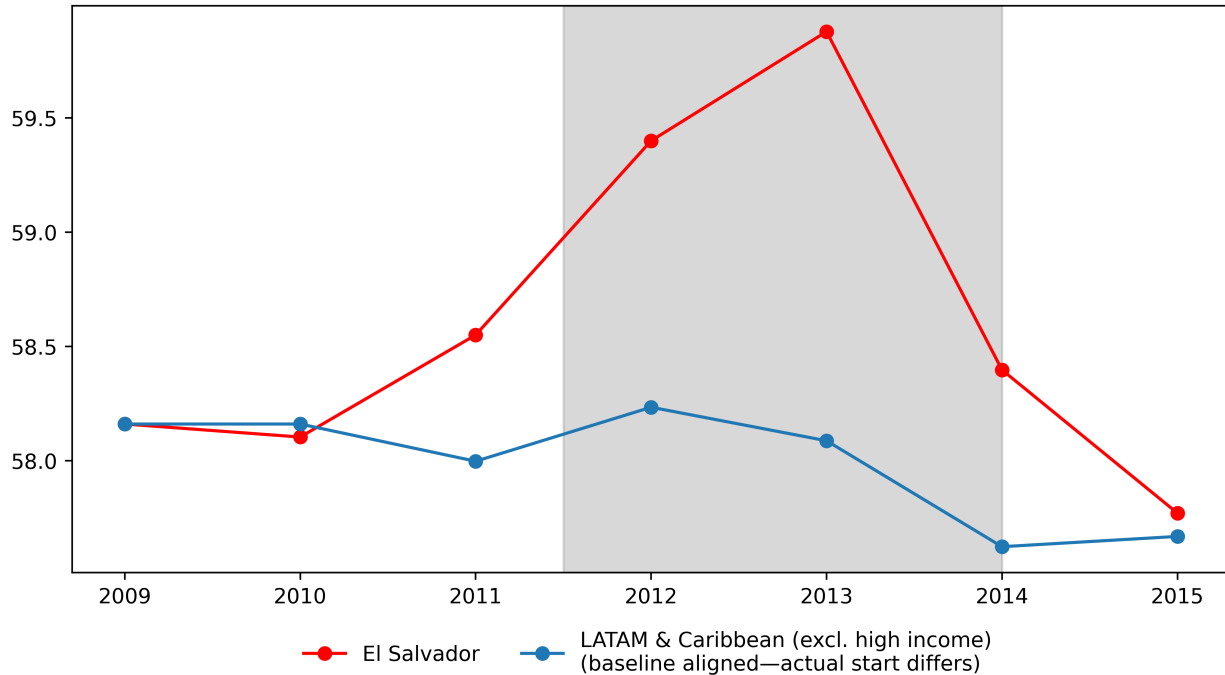
Notes: This figure presents the monthly percentage growth of employment (top panel) and firms (bottom panel) over the study period. Each bar represents the month-to-month growth rate, while the dashed red and orange lines indicate the average growth rates for specific periods. The shaded gray area highlights the truce period.

Figure B.4: Trends in Monthly Reported Extortions During the Truce Period



Notes: This graph illustrates the trend in monthly reported extortions from 2011 to 2017, with a specific focus on the truce period highlighted in grey. Despite the implementation of the truce, aimed at reducing gang violence, the data indicates that extortion rates remained relatively stable throughout the truce period.

Figure B.5: Employment to Population Ratio (%) Over Time Using WDI Data



Notes: This figure presents a key labor market indicator for the period 2009–2015, using data from the World Development Indicators (WDI), which are based on household surveys. The graph displays the employment-to-population ratio (15+, total %), illustrating the proportion of the working-age population engaged in employment each year. The shaded area corresponds to the period of the gang truce (2012–2014). For comparison, the figure also includes the regional trend for Latin America and the Caribbean (excluding high-income countries), baseline-aligned to El Salvador’s truce period.

Table B.1: Employment to Population Ratio Over Time Using WDI Data

Region	Year-by-Year							Summary		
	2009	2010	2011	2012	2013	2014	2015	Pre-Truce	Truce	Difference
El Salvador	58.16	58.10	58.55	59.40	59.88	58.40	57.77	58.27	59.22	0.95
LATAM and Caribbean (excl. high income)	60.16	60.16	60.00	60.24	60.09	59.63	59.67	60.11	59.99	-0.12
Difference	-2.00	-2.06	-1.45	-0.84	-0.21	-1.23	-1.90	-1.84	-0.76	1.08

Notes: This table presents employment rate for the period 2009–2015, using data from the World Development Indicators (WDI), which are based on household surveys. The employment-to-population ratio (15+, total %), is the proportion of the working-age population engaged in employment each year. The truce period spans 2012–2014, and the pre-truce period corresponds to the earlier years shown in the table (2009–2011).

Table B.2: Summary Statistics: Treatment and $NN(10)$ Matched Control Areas with a 0.3-SD Caliper and 2 Reuses

Variable	Matched Control		Treatment	
	Mean	SD	Mean	SD
<i>Panel A</i>				
Number of firms per polygon	11.84	17.99	13.48	17.62
Number of firms per polygon (excl. top 5%)	11.33	17.11	12.91	16.88
Number of workers per polygon	173.56	428.32	200.41	388.44
Number of workers per polygon (excl. top 5%)	72.83	121.68	86.51	126.16
Average wages	327.74	202.23	331.71	211.16
Average wages (excl. top 5%)	278.55	94.02	286.95	91.49
Number of opened firms per quarter	0.36	0.97	0.37	0.76
Number of closed firms per quarter	0.21	0.60	0.29	0.61
<i>Panel B</i>				
Homicide rate (per 100k inhabitants) before truce	7.09	19.19	7.83	13.89
Homicide rate (per 100k inhabitants) after truce	4.18	15.50	4.05	10.94
Number of polygons	835.00		225.00	

Notes: Panel A reports descriptive statistics on business activity, while Panel B reports descriptive statistics on violence before and after the truce. The matched-control group corresponds to the specification that uses 10 nearest neighbors, a 0.3-standard-deviation caliper, and allows each control polygon to be reused at most twice.

Table B.3: Summary Statistics: Treatment and $NN(15)$ Matched Control Areas with a 0.25-SD Caliper and 4 Reuses

Variable	Matched Control		Treatment	
	Mean	SD	Mean	SD
<i>Panel A</i>				
Number of firms per polygon	12.08	16.62	13.13	16.31
Number of firms per polygon (excl. top 5%)	11.60	15.93	12.51	15.54
Number of workers per polygon	165.99	359.65	212.07	461.07
Number of workers per polygon (excl. top 5%)	73.92	111.58	83.11	114.45
Average wages	328.54	205.93	337.28	221.69
Average wages (excl. top 5%)	278.54	92.16	288.39	95.93
Number of opened firms per quarter	0.37	0.97	0.40	0.86
Number of closed firms per quarter	0.21	0.56	0.26	0.52
<i>Panel B</i>				
Homicide rate (per 100k inhabitants) before truce	7.06	18.76	7.73	13.82
Homicide rate (per 100k inhabitants) after truce	4.06	14.95	4.04	10.91
Number of polygons	810.00		225.00	

Notes: Panel A reports descriptive statistics on business activity, while Panel B reports descriptive statistics on violence before and after the truce. The matched-control group corresponds to the specification that uses 15 nearest neighbors, a 0.25-standard-deviation caliper, and allows each control polygon to be reused at most four times.

Table B.4: Difference-in-Differences Estimates Across Matched Specifications ($NN(5)$, $NN(10)$, and $NN(15)$)

Variables	Workers			Number of firms		
	K5, c=0.35, r=0	K10, c=0.3, r=2	K15, c=0.25, r=4	K5, c=0.35, r=0	K10, c=0.3, r=2	K15, c=0.25, r=4
Treatment	2.333* (1.341)	2.226 (1.603)	2.766* (1.640)	0.467*** (0.155)	0.529*** (0.182)	0.561*** (0.184)
Mean	71.368	86.750	82.703	11.627	13.612	13.172
R-squared	0.9867	0.9848	0.9832	0.9933	0.9926	0.9916
Obs.	20448	16960	16560	20448	16960	16560
Clusters	1278	1060	1035	1278	1060	1035

Notes: This table compares difference-in-differences estimates for the effect of the truce on the number of workers and firms across alternative matched-control specifications. $NN(k)$ denotes k -nearest-neighbor matching on pre-truce firm counts. The $NN(5)$ design matches without replacement; $NN(10)$ and $NN(15)$ allow limited replacement as described in the corresponding summary-statistics tables. All regressions include unit and time fixed effects. Standard errors are shown in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.5: Matching Balance Summary Across Matched Specifications ($NN(5)$, $NN(10)$, and $NN(15)$)

Spec.	Treated mean	Matched trol mean	con- Standardized mean diff.	Control poly- gons used	Max reuse
K5, c=0.35, r=0	11.479	8.560	0.183	1065	0
K10, c=0.3, r=2	13.458	11.717	0.098	835	2
K15, c=0.25, r=4	13.084	11.956	0.069	810	4

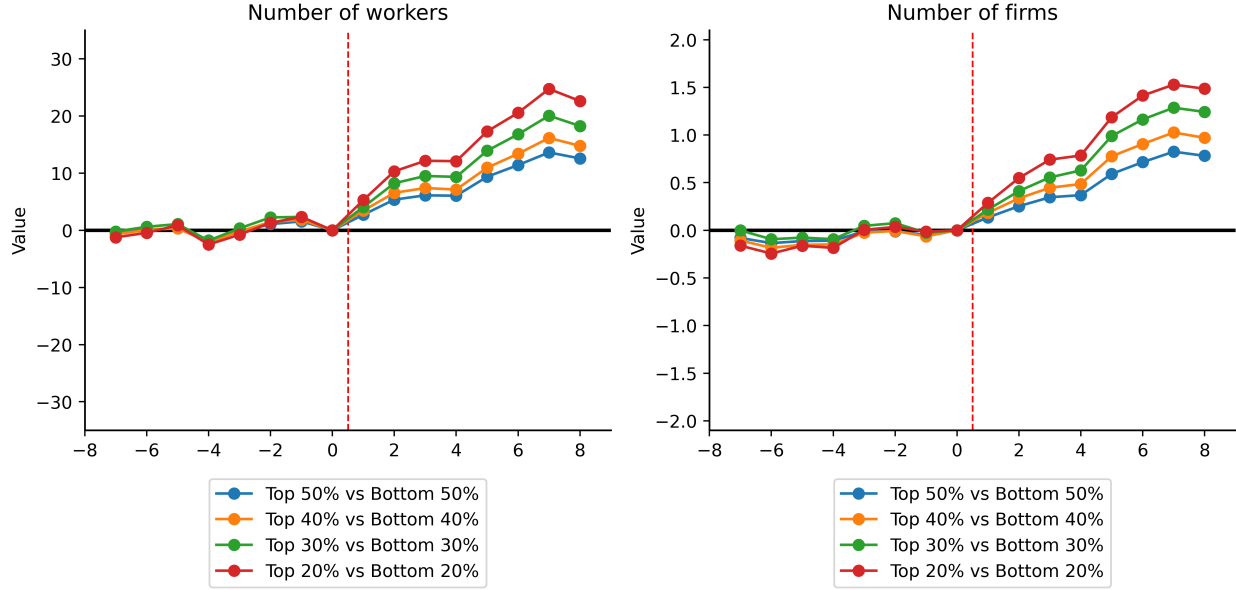
Notes: This table summarizes covariate balance across the matched specifications reported in this appendix. $NN(k)$ denotes k -nearest-neighbor matching on pre-truce firm counts. The table compares the treated-group mean with the corresponding matched-control means and reports standardized differences after matching for the $NN(5)$, $NN(10)$, and $NN(15)$ designs. Lower standardized differences indicate closer balance between treated and matched-control polygons.

Table B.6: Summary Statistics: Treatment and Control Areas by Realized Homicide Reductions

Variable	Bottom 50% homicide reductions		Top 50% homicide reductions	
	Mean	SD	Mean	SD
<i>Panel A</i>				
Number of firms per polygon	8.54	24.67	22.22	82.25
Number of firms per polygon (excl. top 5%)	8.11	22.87	20.91	76.13
Number of workers per polygon	104.71	310.31	253.33	750.98
Number of workers per polygon (excl. top 5%)	55.37	185.2	151.42	627.29
Average wages	556.48	557.16	736.21	551.79
Average wages (excl. top 5%)	500.32	477.5	658.1	447.36
Number of opened firms per quarter	0.26	0.86	0.62	2.66
Number of closed firms per quarter	0.17	0.6	0.43	1.67
<i>Panel B</i>				
Homicide rate (per 100k inhabitants) before truce	4.22	14.98	9.63	21.51
Homicide rate (per 100k inhabitants) after truce	3.93	16.42	4.27	14.89
Number of polygons	807.0		774.0	

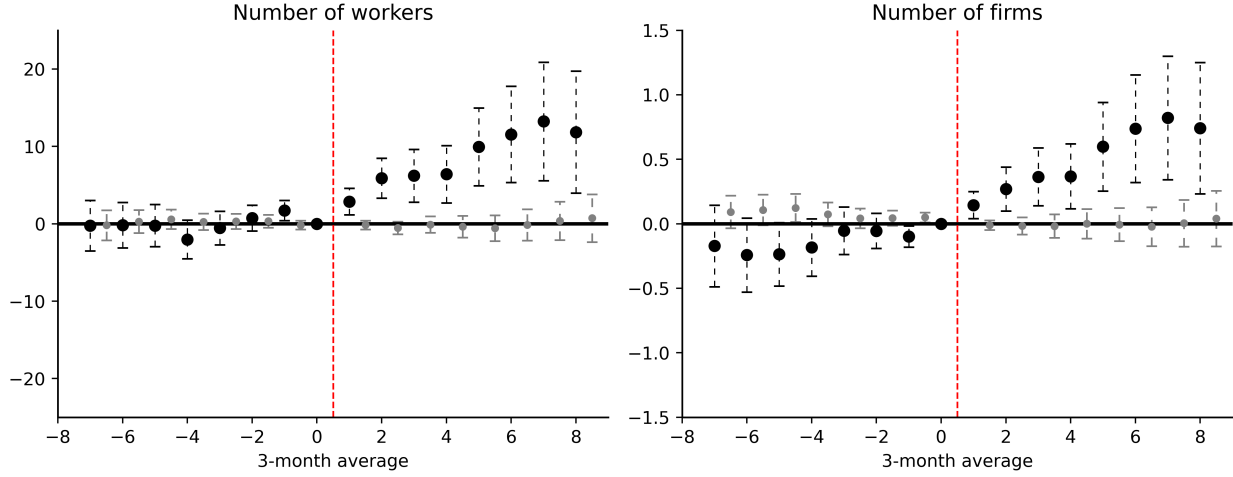
Notes: Treated areas are polygons in the top 50% of the distribution of realized homicide reductions during the truce, while control areas are polygons in the bottom 50%. Panel A reports descriptive statistics on business activity, while Panel B provides descriptive statistics on crime, drawing on homicide rates observed 24 months before and after the truce.

Figure B.6: Event-Study Results for Different Treatment-Intensity Thresholds



Notes: This figure presents event-study estimates based on Equation (1), comparing outcomes across treated units with varying degrees of treatment intensity. Specifically, it plots separate estimates for units in the top 50% vs. bottom 50%, top 40% vs. bottom 40%, top 30% vs. bottom 30%, and top 20% vs. bottom 20% of the homicide reduction distribution during the truce period. The X-axis aggregates data into three-month intervals spanning 24 months before and after the truce. All regressions include unit and time fixed effects.

Figure B.7: Event-Study Results for Placebo Truce Assignments



Notes: This figure presents event-study estimates based on the specification in Equation (6), which compares the estimated treatment effects for the actual truce period versus placebo periods starting 3, 5, 7, and 9 months prior. Each dataset defines treatment based on abnormal homicide reductions, and is stacked with a scenario indicator s identifying the truce or placebo period. Unit and time fixed effects are fully interacted with s . The X-axis consolidates time into three-month intervals spanning 24 months before and after the treatment assignment. Black lines represent the estimated effects for the actual truce, while green lines display the average placebo effects across all alternative scenarios. Vertical bars represent 95% confidence intervals, with standard errors clustered at the unit $\times$ scenario level.

Table B.7: Sensitivity of Treatment Effects to Alternative Sample Cutoffs

Variables	Workers		Number of firms	
	Coef/SE	Per. Change	Coef/SE	Per. Change
<i>Top 50% vs Bottom 50%</i>				
Treatment	8.313*** (2.537)	(5.490%)	0.566*** (0.184)	(2.544%)
R-squared	0.9958		0.9986	
Obs.	25296		25296	
<i>Top 40% vs Bottom 40%</i>				
Treatment	9.884*** (3.130)	(5.666%)	0.727*** (0.225)	(2.854%)
R-squared	0.9959		0.9987	
Obs.	19808		19808	
<i>Top 30% vs Bottom 30%</i>				
Treatment	11.939*** (4.053)	(5.778%)	0.833*** (0.289)	(2.785%)
R-squared	0.9961		0.9987	
Obs.	14848		14848	
<i>Top 20% vs Bottom 20%</i>				
Treatment	15.681*** (5.814)	(6.380%)	1.088*** (0.395)	(3.083%)
R-squared	0.9962		0.9989	
Obs.	9920		9920	
<i>Top 10% vs Bottom 10%</i>				
Treatment	9.997** (4.297)	(5.898%)	0.677 (0.415)	(2.564%)
R-squared	0.993		0.9965	
Obs.	5104		5104	

Notes: This table reports difference-in-differences estimates for the effect of the truce on the number of workers and firms, using a two-way fixed effects specification analogous to Equation (1). Each column corresponds to a different treatment cutoff, comparing units in the top versus bottom $x\%$ of homicide reductions during the truce period (e.g., 50% vs 50%, 40% vs 40%, etc.). The sample excludes the top 5% of firms by size. All regressions include unit and time fixed effects. Percentage changes (Per. Change), shown in parentheses, are computed relative to the pre-truce baseline mean. Standard errors are clustered at the level of geographic units. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.8: Placebo Triple-Differences Estimates

Variables	3 months be- fore	5 months be- fore	7 months be- fore	9 months be- fore	All placebos
<i>Workers</i>					
Treatment (Placebo)	-0.387 (2.060)	-0.112 (1.841)	0.038 (1.610)	-0.641 (1.394)	-0.275 (0.872)
Treatment x Truce	8.700*** (3.267)	8.424*** (3.134)	8.274*** (3.004)	8.953*** (2.894)	8.588*** (2.682)
R-squared	0.9961	0.9964	0.9965	0.9966	0.9968
Obs.	50592	50592	50592	50592	126480
<i>Number of firms</i>					
Treatment (Placebo)	0.102 (0.170)	-0.146 (0.160)	-0.129 (0.155)	-0.108 (0.148)	-0.070 (0.079)
Treatment x Truce	0.463* (0.250)	0.711*** (0.244)	0.694*** (0.241)	0.674*** (0.236)	0.636*** (0.200)
R-squared	0.9987	0.9987	0.9988	0.9988	0.9988
Obs.	50592	50592	50592	50592	126480

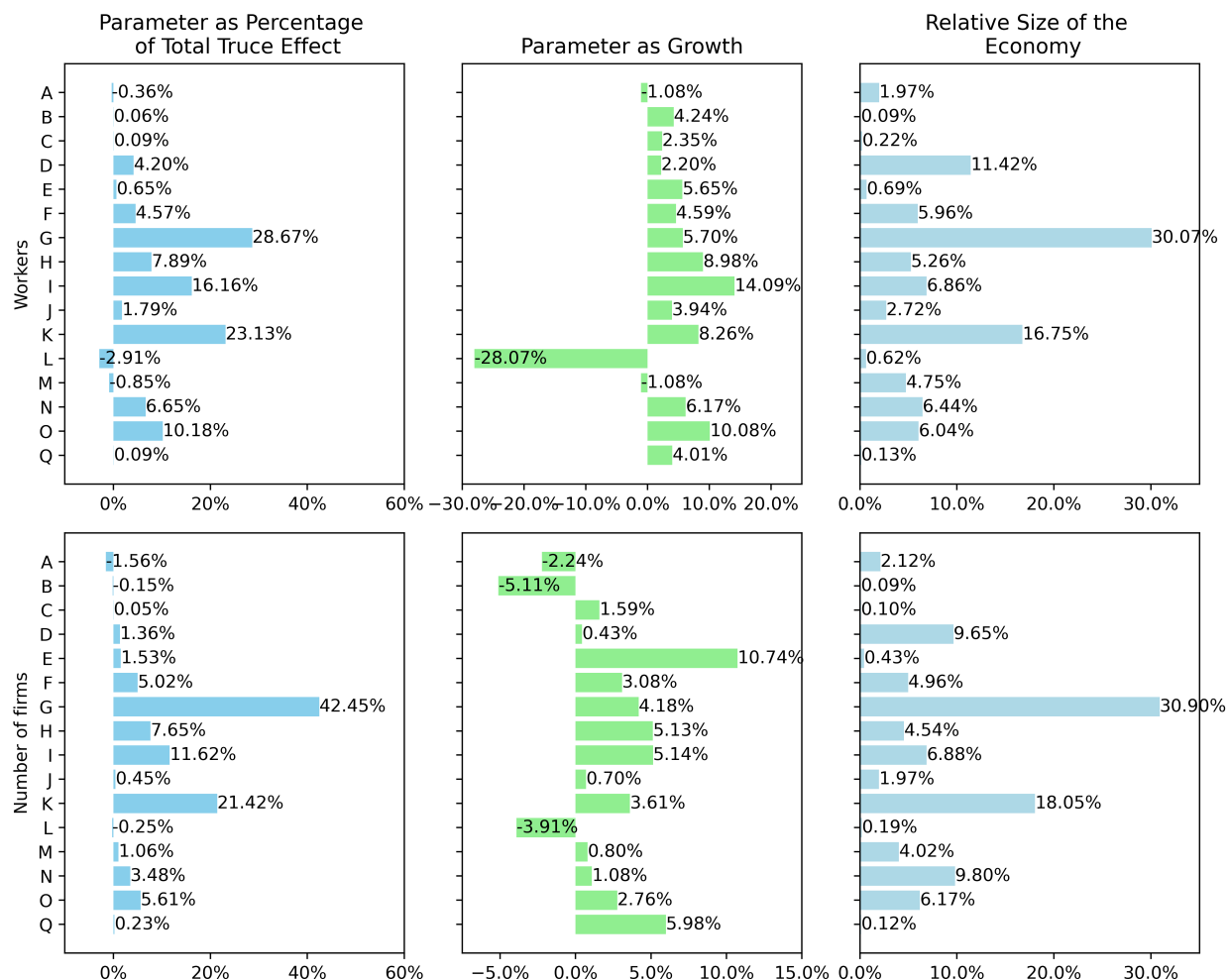
Notes: This table reports the triple-differences estimates examining the effect of the truce and several placebo periods set 3, 5, 7, and 9 months prior to the actual onset of the truce. Each column presents results from a separate specification. The rows labeled “Treatment (Placebo)” capture the hypothetical impact of assigning treatment status using the same procedure but well before the truce actually occurred. The row “Treatment x Truce” represents the additional effect observed exclusively during the actual truce period. Standard errors (in parentheses) are clustered at the polygon level interacted with placebo period or truce. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table B.9: Placebo Triple-Differences Estimates by Treatment Intensity

Variables	Treatment	Lower Threshold	Middle Threshold	Upper Threshold
<i>Workers</i>				
Treatment (Placebo)	-0.275 (0.872)	0.363 (0.952)	1.830 (1.142)	3.638*** (1.389)
Treatment x Truce	8.588*** (2.682)	10.573*** (3.386)	12.239** (4.787)	16.613* (8.862)
R-squared	0.9968	0.9968	0.997	0.9968
Obs.	126480	111024	95616	80256
<i>Number of firms</i>				
Treatment (Placebo)	-0.070 (0.079)	-0.011 (0.085)	0.104 (0.101)	0.164 (0.124)
Treatment x Truce	0.636*** (0.200)	0.814*** (0.244)	0.921*** (0.333)	1.283** (0.559)
R-squared	0.9988	0.9989	0.9989	0.9989
Obs.	126480	111024	95616	80256

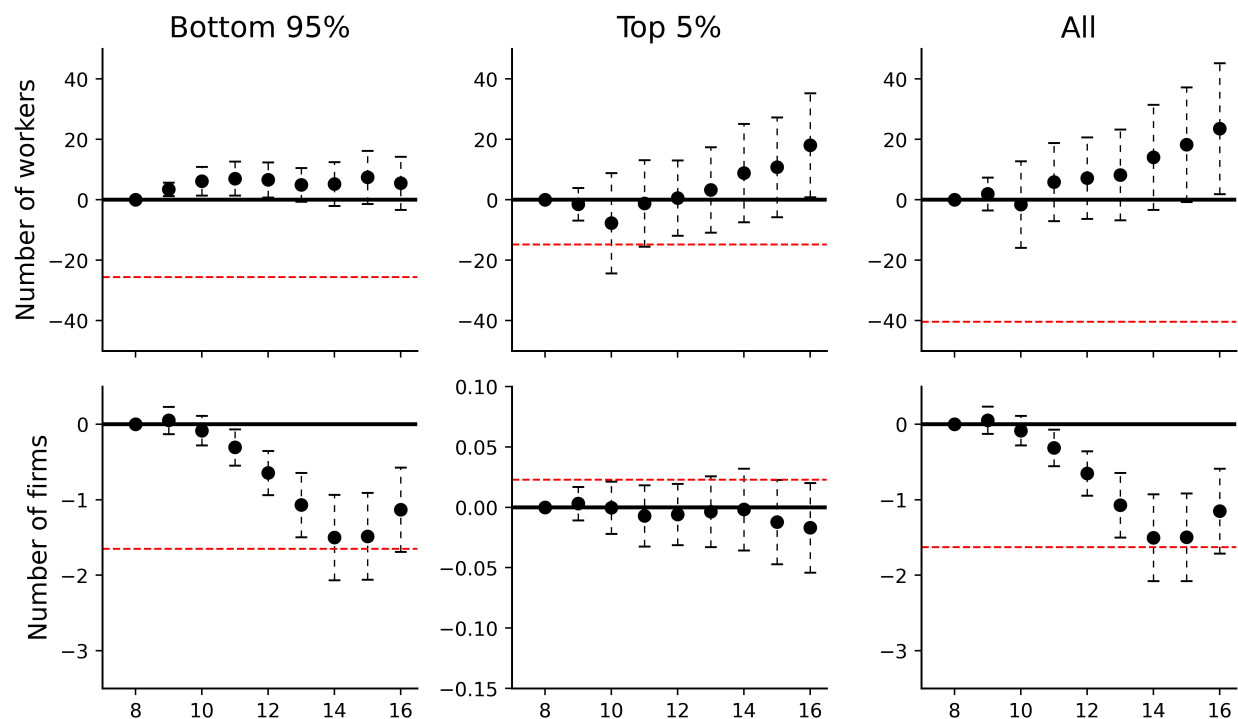
*Notes: This table reports the triple-differences estimates examining the effect of the truce and several placebo periods set 3, 5, 7, and 9 months prior to the actual onset of the truce. Each column presents results from a separate specification. The “Treatment” column presents the baseline top-half homicide-reduction split, while the lower, middle, and upper thresholds apply the same cutoffs used in the threshold-specific exercises, defined by the 25th, 50th, and 75th percentiles of the homicide-reduction distribution among treated polygons. The rows labeled “Treatment (Placebo)” capture the hypothetical impact of assigning treatment status using the same procedure but well before the truce actually occurred. The row “Treatment x Truce” represents the additional effect observed exclusively during the actual truce period. Standard errors (in parentheses) are clustered at the polygon level interacted with placebo period or truce. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$*

Figure B.8: Sectoral Growth and Economic Proportions During the Truce



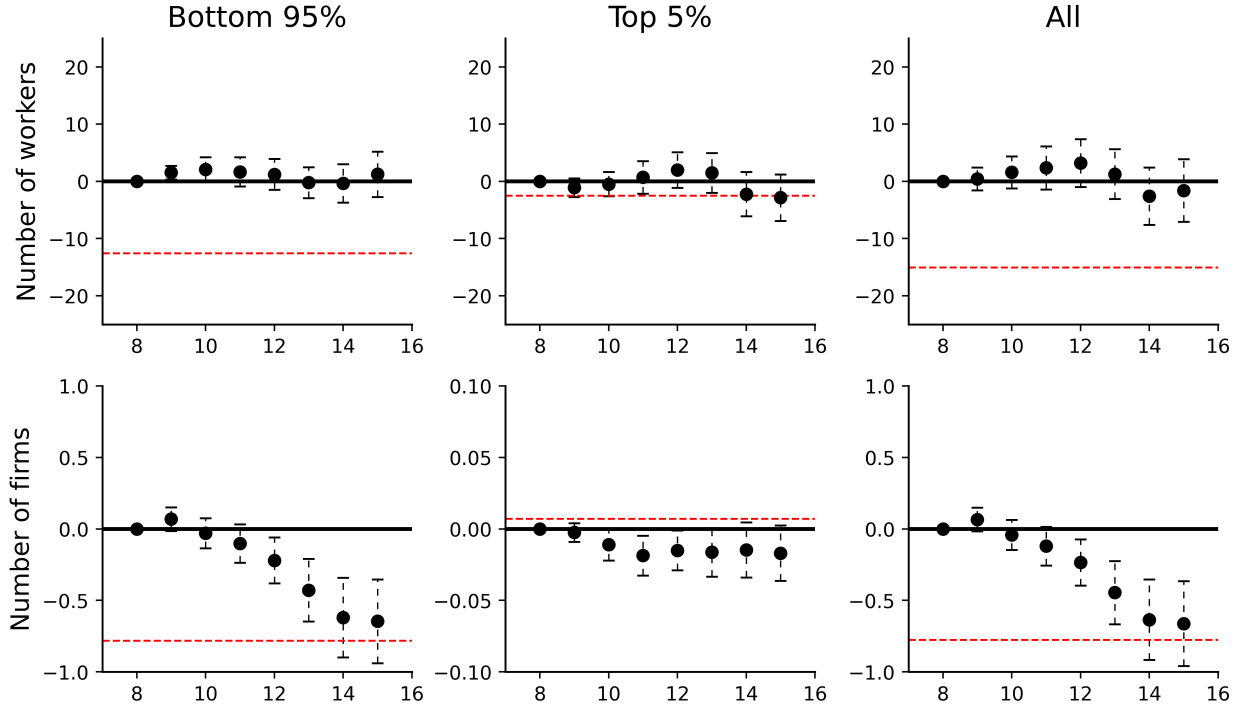
Notes: This graph presents a three-part analysis based on different sectors as classified by the CIIU Rev. 3 standard. The first column represents baseline regression estimates, calculated using sector-specific outcome variables to assess the impact of the truce on each sector. The second column indicates the relative growth in each sector due to the truce, calculated as the treatment effect estimator divided by the pre-truce levels of the corresponding indicator. The third column provides context by showing the percentage that each sector contributes to the overall economy just prior to the truce. The sample excludes the top 5% of firms.

Figure B.9: Post-Truce Event-Study Estimates by Firm Size: Measured Gang Exposure



Notes: This figure presents event-study estimates following the end of the gang truce, using a specification analogous to Equation (1). Exposure is defined using measured pre-truce gang-related homicide activity. Outcomes are shown separately for small and medium firms (left panel), large firms (center panel), and all firms combined (right panel). Time is aggregated into three-month intervals after the end of the truce. The red dotted horizontal line represents the estimated outcome difference in the pre-truce reference period. All regressions include unit and time fixed effects. Vertical bars represent 95% confidence intervals, with standard errors clustered at the geographic unit level.

Figure B.10: Post-Truce Event-Study Estimates by Firm Size: Homicide-Reduction Classification



Notes: This figure presents event-study estimates following the end of the gang truce, using the complementary homicide-reduction classification from the previous specification. The classification compares polygons in the top 50% of realized homicide reductions during the truce to polygons in the bottom 50%. Outcomes are shown separately for small and medium firms (left panel), large firms (center panel), and all firms combined (right panel). Time is aggregated into three-month intervals spanning 24 months after the truce. The red dotted horizontal line represents the average level of the outcome variable immediately prior to the truce. All regressions include unit and time fixed effects. Vertical bars represent 95% confidence intervals, with standard errors clustered at the geographic unit level.

Figure B.11: Crime, Public Perceptions, and Consumer Confidence During the Truce and Non-Aggression Pact Periods



Notes: This figure shows descriptive evidence related to the 2012 gang truce and subsequent non-aggression pact periods in El Salvador. **Panel A** displays the national monthly homicide rate (per 100,000 inhabitants), aggregated from administrative data and smoothed with a 3-month moving average. **Panel B** illustrates the distribution of survey responses to a question on the most pressing problems facing the country, using LAPOP nationally representative household data. **Panel C** shows the evolution of the Consumer Confidence Index. The shaded gray areas indicate two key episodes: the gang truce (March 2012) and the non-aggression pact (April 2016).

B.4 Theoretical Framework

This appendix develops a parsimonious model that organizes the mechanisms suggested by the data. The economy is static, competitive, and features two geographic zones $z \in \{H, L\}$ producing a local non-tradable. For clarity in the exposition, whenever referring to the *low-crime zone* in the text I write *zone L*; symbols with subscript L (e.g., q_L, P_L) remain as standard notation.

The setup features two deviations from a standard local-demand framework: (i) crime enters the *consumer side* as a fear-cost wedge that raises the effective price of purchasing locally in the high-crime zone H ; and (ii) crime also shifts the *labor-supply side* by lowering participation at a given wage (crime-sensitive participation margin).

B.4.1 Environment and Agents

Time is static. There is a representative household of unit mass and a competitive sector of firms in each zone. Labor is perfectly mobile across zones and the production technology is linear in labor with common productivity $A > 0$.

Zones and crime. I index zones by $z \in \{H, L\}$ and let $c \geq 0$ denote the (exogenous) crime level specific to the high-crime zone H . For parsimony I normalize crime in the low-crime zone L to zero; allowing $c_L > 0$ is a straightforward variation.

B.4.2 Firms and Technology

Firms in each zone hire local labor ℓ_z and produce local output y_z using

$$y_z = A \ell_z, \quad A > 0.$$

Perfect competition and free mobility of labor imply that the (common) producer price equals unit cost,

$$p = \frac{w}{A},$$

where w is the common market wage. Local goods are non-tradable, so market clearing in each zone requires $y_z = q_z$, equivalently $A \ell_z = q_z$.

B.4.3 Goods, Crime, and Effective Prices (numéraire iceberg; household closure)

Crime affects consumption in H via a reduced-form fear-cost wedge $\chi(c)$ that raises the *effective consumer price* paid by households:

$$P_L \equiv p, \quad P_H \equiv p + \chi(c), \quad \chi(c) = \beta c, \quad \beta > 0.$$

Household income, utility, and budget. Preferences are quasilinear in the numéraire:

$$U = a_H \ln q_H + a_L \ln q_L + m, \quad a_H, a_L > 0.$$

Total numéraire income is

$$Y \equiv \bar{Y} + wL, \quad \bar{Y} \geq 0,$$

where $\bar{Y}$ is an exogenous endowment/transfer and wL is labor income (firms earn zero profits under perfect competition and $y_z = A\ell_z$). The representative household chooses (q_H, q_L, m) subject to

$$m + P_H q_H + P_L q_L = Y.$$

Resource accounting (numéraire sink). Of the expenditure $P_H q_H$, firms receive $p q_H$; the residual $\chi(c) q_H$ is numéraire *used and destroyed* (no agent receives it). Aggregating,

$$\bar{Y} + wL = m + p(q_H + q_L) + \chi(c) q_H,$$

and with $p = \frac{w}{A}$ and local clearing $q_z = A\ell_z$, we have $p(q_H + q_L) = wL$, so the accounting identity holds.

Feasibility (numéraire nonnegativity). Using $q_H^* = \frac{a_H}{p^* + \chi(c)} = \frac{a_H A}{w^* + A\beta c}$,

$$m^* = \bar{Y} - \chi(c) q_H^* = \bar{Y} - \beta c \cdot \frac{a_H A}{w^* + A\beta c}.$$

Thus the interior allocation is feasible provided

$$\boxed{\bar{Y} \geq \chi(c) q_H^* = \beta c \cdot \frac{a_H A}{w^* + A\beta c}}.$$

In what follows we assume $\bar{Y}$ is large enough so that $m^* \geq 0$. Because preferences are quasilinear, q_H and q_L depend only on (p, χ) ; m adjusts residually.

Demand. Utility maximization yields the Marshallian demands used throughout:

$$q_H = \frac{a_H}{p + \chi(c)}, \quad q_L = \frac{a_L}{p}.$$

Remark on “closed economy.” “Closed” here refers to the local non-tradable good and internal wage/price determination: goods and labor markets clear within the economy. The numéraire is modeled as an endowment/transfer $\bar{Y}$ financing residual consumption m .

B.4.4 Crime-Sensitive Supply, Labor Demand, and Interior Branch

Labor Supply. Crime also affects participation. I model this with a linear shift in the labor-supply schedule:

$$L^S(w, c) = s[w - \psi(c)]_+, \quad \psi(c) \equiv \gamma c, \quad s > 0, \gamma > 0,$$

where $[x]_+ = \max\{x, 0\}$. Parameter s is the slope (extensive-margin elasticity) of labor supply and γ governs how much crime lowers willingness to work (commuting risk, time losses, safety concerns).

Labor demand. Using $A\ell_z = q_z$, $p = w/A$, and $\chi(c) = \beta c$, the zone-level and aggregate labor demands are

$$\ell_H(w, c) = \frac{a_H}{w + A\beta c}, \quad \ell_L(w) = \frac{a_L}{w}, \quad L^D(w, c; a_H, a_L) = \frac{a_H}{w + A\beta c} + \frac{a_L}{w}.$$

Interior branch. For any $w \in (0, \gamma c]$, $L^S(w, c) = 0$ while $L^D(w, c; a_H, a_L) > 0$. At the participation boundary, $L^D(\gamma c, c) \in (0, \infty)$ if $c > 0$; and for any $c \geq 0$, $\lim_{w \downarrow 0} L^D(w, c) = +\infty$. Hence the market-clearing condition $L^S = L^D$ cannot hold at or below the participation threshold, and any equilibrium must satisfy $w^* > \gamma c$. On the interior domain $(\gamma c, \infty)$, $L^S(w, c) = s(w - \gamma c)$ is differentiable and strictly increasing, which I use in the existence and uniqueness proof below.

B.5 Competitive equilibrium: definition, existence, and uniqueness

I now characterize the competitive equilibrium for the baseline specification with size-only local demand and crime-sensitive labor supply introduced above.

B.5.1 Definition of equilibrium

Given parameters $(A, a_H, a_L, s, \beta, \gamma)$ and the crime level $c \geq 0$, a (competitive) equilibrium is a wage $w^* > 0$ and associated allocations

$$\begin{aligned} p^* &= \frac{w^*}{A}, & P_H^* &= p^* + \beta c, & P_L^* &= p^*, \\ q_H^* &= \frac{a_H}{P_H^*} = \frac{a_H A}{w^* + A\beta c}, & q_L^* &= \frac{a_L}{P_L^*} = \frac{a_L A}{w^*}, \\ \ell_H^* &= \frac{q_H^*}{A} = \frac{a_H}{w^* + A\beta c}, & \ell_L^* &= \frac{q_L^*}{A} = \frac{a_L}{w^*}, \end{aligned}$$

such that local non-tradable markets clear ($y_z^* = q_z^*$) and aggregate labor supply equals aggregate labor demand:

$$\begin{aligned} L^S(w^*, c) &= L^D(w^*, c; a_H, a_L), \\ L^S(w, c) &= s[w - \gamma c]_+, & L^D(w, c; a_H, a_L) &= \frac{a_H}{w + A\beta c} + \frac{a_L}{w}. \end{aligned} \tag{E}$$

I focus on the empirically relevant interior case $w^* > \gamma c$, under which $L^S(w, c) = s(w - \gamma c)$ is differentiable.

B.5.2 Existence and uniqueness (baseline size-only)

Lemma B.5.1 (Properties and boundary behavior of L^D) *For any $a_H, a_L, A > 0, \beta > 0$, and finite $c \geq 0$, the aggregate labor demand*

$$L^D(w, c; a_H, a_L) = \frac{a_H}{w + A\beta c} + \frac{a_L}{w}$$

is continuous on $(0, \infty)$, strictly decreasing in w , and satisfies

$$\lim_{w \downarrow 0} L^D(w, c) = +\infty, \quad \lim_{w \uparrow \infty} L^D(w, c) = 0, \quad \frac{\partial L^D}{\partial w}(w, c) = -\frac{a_H}{(w + A\beta c)^2} - \frac{a_L}{w^2} < 0.$$

Moreover, at the participation boundary $w = \gamma c$:

$$\begin{cases} L^D(\gamma c, c) = \frac{a_H}{(\gamma + A\beta)c} + \frac{a_L}{\gamma c} \in (0, \infty), & \text{if } c > 0, \\ \lim_{w \downarrow 0} L^D(w, 0) = +\infty, & \text{if } c = 0. \end{cases}$$

Proof. All statements follow directly from the closed form and elementary limits. ■

Lemma B.5.2 (Properties of L^S on the interior branch) For $w > \gamma c$, $L^S(w, c) = s(w - \gamma c)$ is continuous and strictly increasing in w , with $L^S(\gamma c, c) = 0$ and $\lim_{w \uparrow \infty} L^S(w, c) = +\infty$.

Proof. Immediate from linearity with $s > 0$. ■

Proposition 1 (Existence and uniqueness) Fix $a_H, a_L, A, s > 0$, $\beta, \gamma > 0$, and finite $c \geq 0$. There exists a unique $w^* > \gamma c$ solving (E).

Proof. Let $H(w; c) \equiv L^D(w, c; a_H, a_L) - L^S(w, c)$ for $w \geq \gamma c$.

Case $c > 0$. At $w = \gamma c$,

$$L^S(\gamma c, c) = 0, \quad L^D(\gamma c, c) = \frac{a_H}{(\gamma + A\beta)c} + \frac{a_L}{\gamma c} > 0,$$

so $H(\gamma c; c) > 0$.

Case $c = 0$. For $w > 0$, $H(w; 0)$ is continuous and

$$\lim_{w \downarrow 0} H(w; 0) = \lim_{w \downarrow 0} \left(\frac{a_H}{w} + \frac{a_L}{w} - s w \right) = +\infty,$$

since $L^D(w, 0) = (a_H + a_L)/w \rightarrow +\infty$ and $L^S(w, 0) = s w \rightarrow 0$.

In both cases H is continuous on $(\gamma c, \infty)$ and

$$\lim_{w \rightarrow \infty} H(w; c) = \lim_{w \rightarrow \infty} (L^D(w, c) - L^S(w, c)) = 0 - (+\infty) = -\infty.$$

By the Intermediate Value Theorem, there exists $w^* > \gamma c$ with $H(w^*; c) = 0$ (existence). Uniqueness follows because $L^D(\cdot, c)$ is strictly decreasing and $L^S(\cdot, c)$ is strictly increasing on $(\gamma c, \infty)$, so the two schedules can cross at most once. ■

Interiority and feasibility. Proposition 1 implies $w^* > \gamma c$, i.e., the interior branch is the relevant equilibrium (the kink is not binding). Given quasilinear preferences, the Marshallian quantities are income-independent; I assume the numéraire endowment Y is sufficiently large so that the implied $m^* \geq 0$ (feasibility of the interior allocation).

Equilibrium objects (summary). The unique equilibrium is characterized by the scalar w^* that solves (E). All remaining prices and allocations follow from

$$p^* = \frac{w^*}{A}, \quad P_H^* = p^* + \beta c, \quad P_L^* = p^*, \quad (q_H^*, q_L^*) = \left(\frac{a_H A}{w^* + A\beta c}, \frac{a_L A}{w^*} \right), \quad (\ell_H^*, \ell_L^*) = \left(\frac{a_H}{w^* + A\beta c}, \frac{a_L}{w^*} \right).$$

These expressions will be used below to derive the comparative statics with respect to crime.

B.6 Crime and equilibrium outcomes: wage and aggregate employment

We now study how the competitive equilibrium reacts to changes in crime c in the high-crime zone H , focusing on the equilibrium wage $w^*(c)$ and aggregate employment $L^*(c)$. Throughout we retain the size-only demand and crime-sensitive labor supply of (E), and the interior branch $w^* > \gamma c$.

B.6.1 Setup for comparative statics

Let

$$F(w, c) \equiv s[w - \gamma c] - \frac{a_H}{w + A\beta c} - \frac{a_L}{w},$$

so that the equilibrium condition is $F(w^*(c), c) = 0$. Define the convenient shorthands

$$X \equiv w^*(c) + A\beta c, \quad Z \equiv w^*(c).$$

Direct differentiation yields

$$F_w = s + \frac{a_H}{X^2} + \frac{a_L}{Z^2} > 0, \quad F_c = -s\gamma + \frac{a_H A\beta}{X^2}.$$

By the Implicit Function Theorem,

$$\frac{dw^*}{dc} = -\frac{F_c}{F_w} = -\frac{-s\gamma + \frac{a_H A\beta}{X^2}}{s + \frac{a_H}{X^2} + \frac{a_L}{Z^2}}. \quad (7)$$

B.6.2 Equilibrium wage

Proposition 2 (Wage response to crime) *Under the baseline specification, the equilibrium wage is (locally) decreasing in crime if and only if*

$$\frac{a_H A \beta}{X^2} > s \gamma. \quad (C1)$$

Equivalently, when (C1) holds, a reduction in crime ($\Delta c < 0$) raises the wage ($\Delta w^ > 0$).*

Interpretation. Condition (C1) compares the *demand-side* impact of crime (the fear-cost wedge in H , scaled by its market size a_H and productivity A) to the *supply-side* inward shift (crime-sensitive participation $s\gamma$). If the demand effect dominates, crime reductions push the wage up; otherwise the wage response can be attenuated or (locally) flat/negative.

B.6.3 Aggregate employment

Aggregate employment equals

$$L^*(c) = L^S(w^*(c), c) = s[w^*(c) - \gamma c].$$

Differentiating and using (7) gives

$$\begin{aligned} \frac{dL^*}{dc} &= s \left(\frac{dw^*}{dc} - \gamma \right) = s \frac{-\left(-s\gamma + \frac{a_H A \beta}{X^2}\right) - \gamma \left(s + \frac{a_H}{X^2} + \frac{a_L}{Z^2}\right)}{s + \frac{a_H}{X^2} + \frac{a_L}{Z^2}} \\ &= -s \frac{a_H \frac{A\beta + \gamma}{X^2} + a_L \frac{\gamma}{Z^2}}{s + \frac{a_H}{X^2} + \frac{a_L}{Z^2}}. \end{aligned}$$

Proposition 3 (Crime strictly lowers aggregate employment) *Fix $a_H, a_L, s, A > 0$ with $\beta > 0$ and $\gamma > 0$. Then*

$$\frac{dL^*}{dc} < 0.$$

Equivalently, any reduction in crime ($\Delta c < 0$) strictly increases aggregate employment ($\Delta L^ > 0$).*

Intuition. A crime reduction shifts *demand* outward by lowering the effective price in H and shifts *supply* outward by raising participation. Even if the wage response is muted (e.g., supply expands strongly), the outward movements ensure that total employment rises unambiguously.

B.7 Zone-by-zone effects: substitution across zones vs. participation gains

We now decompose the employment response by zone when crime c changes in H . Recall (size-only demand, crime-sensitive supply):

$$\ell_H = \frac{a_H}{w + A\beta c}, \quad \ell_L = \frac{a_L}{w}, \quad L^* = s[w^*(c) - \gamma c],$$

and define the shorthands

$$X \equiv w^* + A\beta c, \quad Z \equiv w^*.$$

Local derivatives and a convenient decomposition. Total derivatives with respect to c can be written as

$$\frac{d\ell_H}{dc} = \underbrace{\left(-\frac{a_H A\beta}{X^2}\right)}_{\text{direct demand (crime wedge in } H) \text{ at fixed } w} + \underbrace{\left(-\frac{a_H}{X^2}\right) \frac{dw^*}{dc}}_{\text{wage (general-equilibrium) channel}}, \quad (8)$$

$$\frac{d\ell_L}{dc} = \underbrace{0}_{\text{no direct crime term in the low-crime zone}} + \underbrace{\left(-\frac{a_L}{Z^2}\right) \frac{dw^*}{dc}}_{\text{wage (general-equilibrium) channel}}. \quad (9)$$

Thus, any change in employment in the low-crime zone is *purely* a reallocation (via the wage), while the high-crime zone combines a direct local-demand force with the wage channel.

Using the equilibrium condition (Section B.5), the wage derivative is

$$\frac{dw^*}{dc} = -\frac{-s\gamma + \frac{a_H A\beta}{X^2}}{s + \frac{a_H}{X^2} + \frac{a_L}{Z^2}} \equiv -\frac{F_c}{F_w}, \quad F_w > 0. \quad (10)$$

B.7.1 When does cross-zone substitution occur?

A *crime reduction* ($\Delta c < 0$) raises aggregate employment L^* (Proposition 3). Whether we also observe reallocation away from the low-crime zone L hinges on the wage response.

Low-crime zone monotonicity. Employment in the low-crime zone is $\ell_L(w) = a_L/w$, which is strictly decreasing in w . Thus, for any finite change from c to c' with $w_1 = w^*(c) > 0$ and $w_2 = w^*(c') > 0$,

$$\Delta \ell_L = \ell_L(w_2) - \ell_L(w_1) = a_L \left(\frac{1}{w_2} - \frac{1}{w_1} \right) = -a_L \frac{\Delta w^*}{w_1 w_2}, \quad \Delta w^* \equiv w_2 - w_1,$$

so

$$\text{sign}(\Delta \ell_L) = -\text{sign}(\Delta w^*).$$

Local vs. finite changes. At the margin, reallocation from L occurs iff the wage increases when crime falls:

$$dc < 0 \Rightarrow dw^* > 0 \iff d\ell_L < 0.$$

In our baseline, this local wage increase holds *iff* condition

$$\frac{a_H A \beta}{X^2} > s \gamma \iff \frac{dw^*}{dc} < 0, \quad (\text{C1})$$

with $X \equiv w^* + A\beta c$. For a *finite* crime change Δc , the global implication $\Delta c < 0 \Rightarrow \Delta w^* > 0 \Rightarrow \Delta \ell_L < 0$ follows provided the sign $dw^*/dc < 0$ is maintained along the entire path in c (equivalently, (C1) holds for all intermediate crime levels).

Why ℓ_H must rise. Adding up $L^* = \ell_H + \ell_L$ and using $\Delta L^* > 0$:

- If (C1) holds, then $\Delta \ell_L < 0$ and thus

$$\Delta \ell_H = \Delta L^* - \Delta \ell_L > 0.$$

- If (C1) fails, the wage is weakly non-increasing, so the direct reduction in the H -wedge (lower $A\beta c$ in $w^* + A\beta c$) together with a flat/declining wage implies $\Delta \ell_H > 0$.

Hence, with a crime reduction, employment in H rises unambiguously; whether employment in L falls depends solely on the sign of the wage response via (C1).

Formal proof. Let $X \equiv w^* + A\beta c$ and $Z \equiv w^*$. Zone- H employment is

$$\ell_H = \frac{a_H}{X} \implies \frac{d\ell_H}{dc} = -\frac{a_H}{X^2} \left(\frac{dw^*}{dc} + A\beta \right).$$

Using the equilibrium condition $F(w, c) = 0$ with

$$F_w = s + \frac{a_H}{X^2} + \frac{a_L}{Z^2} > 0, \quad F_c = -s\gamma + \frac{a_H A \beta}{X^2},$$

the Implicit Function Theorem gives

$$\frac{dw^*}{dc} = -\frac{F_c}{F_w} \Rightarrow \frac{dw^*}{dc} + A\beta = \frac{-F_c + A\beta F_w}{F_w} = \frac{s(\gamma + A\beta) + A\beta \frac{a_L}{Z^2}}{F_w} > 0.$$

Therefore $\frac{d\ell_H}{dc} < 0$, i.e., a reduction in crime ($\Delta c < 0$) *strictly* increases employment in H :

$$\Delta c < 0 \Rightarrow \Delta \ell_H > 0.$$

B.7.2 An intuitive reallocation index

To keep the interpretation simple, summarize the composition of the employment gain from a crime reduction ($\Delta c < 0$) with a single *intensity index*:

$$\Delta L^* = \Delta \ell_H + \Delta \ell_L, \quad \text{with } \Delta L^* > 0 \text{ under } \beta, \gamma > 0.$$

Definition (reallocation intensity).

$$\text{Reallocation index } I_{\text{sub}} \equiv \frac{\max\{0, -\Delta \ell_L\}}{\Delta L^*}.$$

When is reallocation active? Reallocation is active ($I_{\text{sub}} > 0$) if and only if the wage rises with the crime reduction, which occurs under the dominance condition

$$\frac{a_H A \beta}{X^2} > s \gamma \iff \Delta w^* > 0 \iff \Delta \ell_L < 0. \quad (\text{C1})$$

How to read the index.

- **Large I_{sub} (reallocation-intensive):** many jobs leave L relative to the net gain. Typical when a_H is large, the fear-cost channel β is strong (or $X = w^* + A\beta c$ is low), and supply sensitivity $s\gamma$ is modest. Empirically: wage up, ℓ_L down.
- **Small I_{sub} (participation-driven):** few jobs leave L compared to the net gain. Typical when s and/or γ are high (wage moves little or falls), or when the base in L is small (a_L small / $Z = w^*$ large). Empirically: wage flat/down, ℓ_L flat/up.